CHAPTER 10

Nitrogen- and Phosphorus-Containing Compounds

Content and rendering review packet

Original app content and actual Chromium screenshots. No textbook passages or textbook figures are embedded. Source labels and curriculum headings are not evidence of factual correctness.

Source snapshot SHA-256: 163a79182f0b65a9a66a5220c65c0d92571f7d4b1d0adb7991e63147fce8aace

Review all claims, molecule graphs, lessons, exercises, hints and displayed states. Keys are author claims, not ground truth. Screenshots show static states with reduced motion: they do not prove animation timing or persistence behavior.

For every finding: item ID/page; severity (critical/major/minor); exact claim; derivation or reference evidence; proposed correction; confidence. List unreviewed/uncertain items. Model agreement is not scientific verification.

Coverage checklist

ch10:exercise:glycine-charge | glycine-charge:stage-1, glycine-charge:stage-2, glycine-charge:stage-3

ch10:exercise:peptide-link | peptide-link:stage-1, peptide-link:stage-2, peptide-link:stage-3

ch10:exercise:strecker-map | strecker-map:stage-1, strecker-map:stage-2, strecker-map:stage-3

ch10:exercise:phosphate-links | phosphate-links:stage-1, phosphate-links:stage-2, phosphate-links:stage-3

ch10:exercise:gabriel-route | gabriel-route:stage-1, gabriel-route:stage-2, gabriel-route:stage-3

ch10:lesson:amino-acids | amino-acids:rule, amino-acids:cue, amino-acids:trap, amino-acids:app, amino-acids:step-1, amino-acids:step-2, amino-acids:step-3

ch10:lesson:peptides | peptides:rule, peptides:cue, peptides:trap, peptides:app, peptides:step-1, peptides:step-2

ch10:lesson:synthesis | synthesis:rule, synthesis:cue, synthesis:trap, synthesis:app, synthesis:step-1, synthesis:step-2, synthesis:step-3

ch10:lesson:phosphorus | phosphorus:rule, phosphorus:cue, phosphorus:trap, phosphorus:app, phosphorus:step-1, phosphorus:step-2

ch10:molecule:alanine | alanine

ch10:molecule:glycine | glycine

ch10:molecule:glycylglycine | glycylglycine

ch10:molecule:aminonitrile | aminonitrile

ch10:molecule:phthalimide-anion | phthalimide-anion

ch10:molecule:dimethylphosphate | dimethylphosphate

ch10:molecule:phosphoricacid | phosphoricacid

ch10:curriculum | scope-1, scope-2, scope-3

Curriculum scope

scope-1: 10.1 Amino Acids, Peptides, and Proteins

scope-2: 10.2 Synthesis of α-Amino Acids

scope-3: 10.3 Phosphorus-Containing Compounds

Objective: Analyze amino-acid charge and stereochemistry

Objective: Identify peptide bonds and synthesis carbon sources

Objective: Distinguish phosphate linkages and reaction energetics

Supplied Kaplan curriculum: Chapter 10; section scope checked against local body and summary | organic_chemistry_kaplan/OEBPS/chapter10b.xhtml

Buffer action and capacity: OpenStax Chemistry 2e 14.6, inspected | https://openstax.org/books/chemistry-2e/pages/14-6-buffers

No source passage is included. Request relevant authoritative excerpts when a claim cannot be checked by derivation.

amino-acids | Amino acids, charge, and stereochemistry

Ionization depends on pH

amino-acids:rule | Core Rules

An alpha amino acid has amino and carboxyl groups attached to the same alpha carbon. At many near-neutral pH conditions, simple amino acids predominantly form zwitterions. Side chains add nonpolar, aromatic, polar, acidic, or basic behavior. Glycine has no stereocenter; most proteinogenic L amino acids are S, with cysteine the familiar R exception. Amino acids are amphoteric: an acidic carboxyl group can donate H+ and a basic amino group can accept H+; side chains may add further ionizable sites.

Schematic summary: Ionization depends on pH

Inventory — Include all ionizable side-chain groups

pH versus pKa — Assign predominant group forms

Sum — Net zero does not mean no formal charges

amino-acids:cue | Memory Cues

Low pH adds protons; high pH removes them. Net charge is the sum of all ionized groups.

Schematic summary: Choose the case that applies

No extra ionizable side chain — Use the pKa pair bracketing the neutral species

Ionizable side chain — Find the neutral species before choosing the pKa pair

amino-acids:trap | Common Mistakes / MCAT Traps

A zwitterion has formal charges despite net zero charge. The pI is the pH of average net zero charge; for ionizable side chains, average the two pKa values bracketing the neutral species, not automatically the terminal pKa pair.

Schematic summary: Shortcut versus a better check

Avoid — A zwitterion has no charges

Check — Positive and negative formal charges can sum to zero

amino-acids:app | MCAT Applications

Aliphatic side chains such as Ala, Val, Leu, and Ile are nonpolar; Ser and Thr have polar OH groups; Asn and Gln have polar amides. Asp/Glu are usually negatively charged near neutral pH; Lys/Arg usually positive. Histidine protonation is strongly pH- and environment-dependent. Aromatic does not always mean purely nonpolar: Tyr has a phenolic OH. At pH below pI a protein tends positive; above pI it tends negative. Methionine has a largely nonpolar thioether side chain; cysteine has a polar thiol and can form disulfides on oxidation. Proline has a cyclic side chain bonded to its nitrogen (a secondary amine in the free amino acid); phenylalanine and tryptophan are commonly grouped as nonpolar aromatic amino acids, although tryptophan has an indole N–H donor.

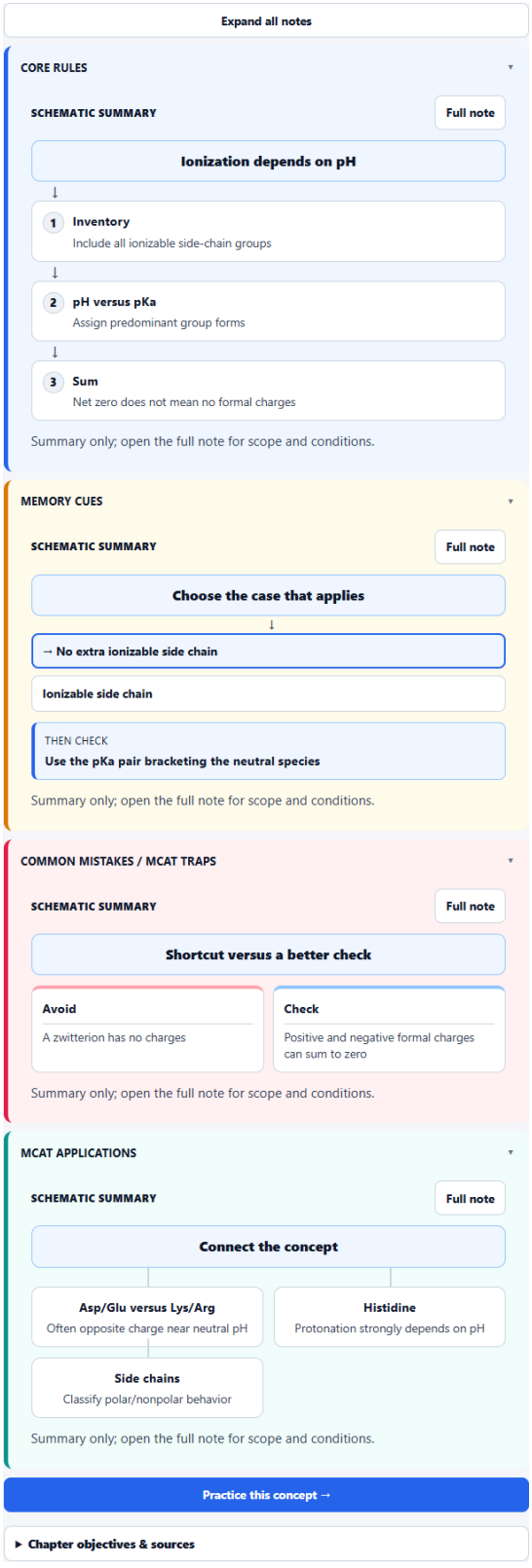
Schematic summary: Connect the concept

Asp/Glu versus Lys/Arg — Often opposite charge near neutral pH

Histidine — Protonation strongly depends on pH

Side chains — Classify polar/nonpolar behavior

Screenshot: amino-acids-visuals



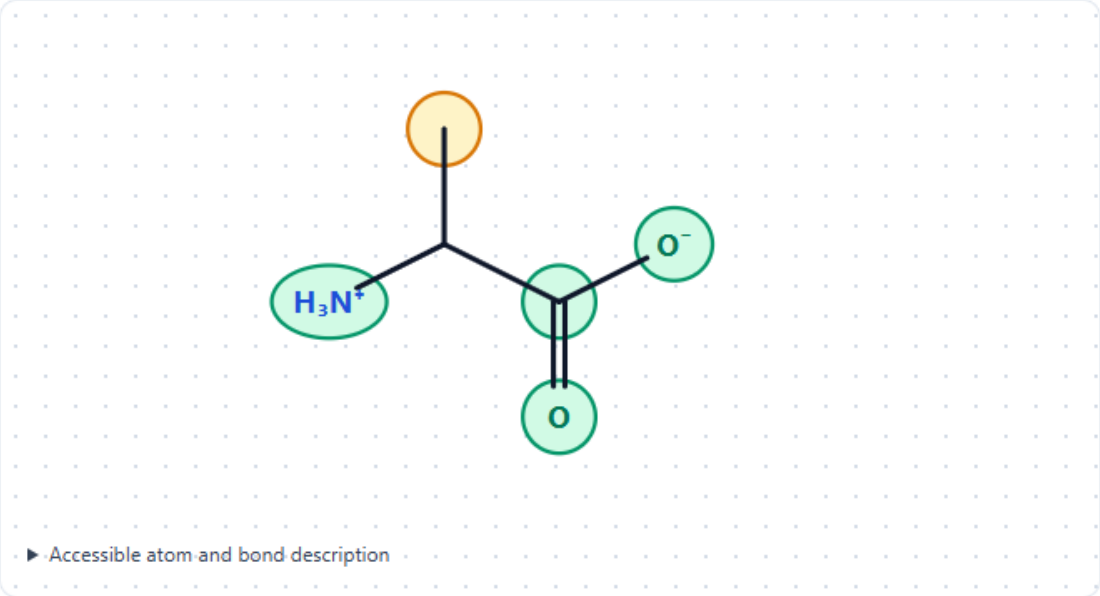
amino-acids:step-1 | 1. Zwitterionic alanine

The nitrogen bears positive charge and the carboxylate bears negative charge. Its alpha carbon also has methyl and H. This connectivity drawing leaves stereochemistry unspecified.

Displayed molecule ID: alanine

{"highlights": {"prin": [0, 2, 3, 4], "sub": [5]}, "revealName": true}

Screenshot: amino-acids-step-1



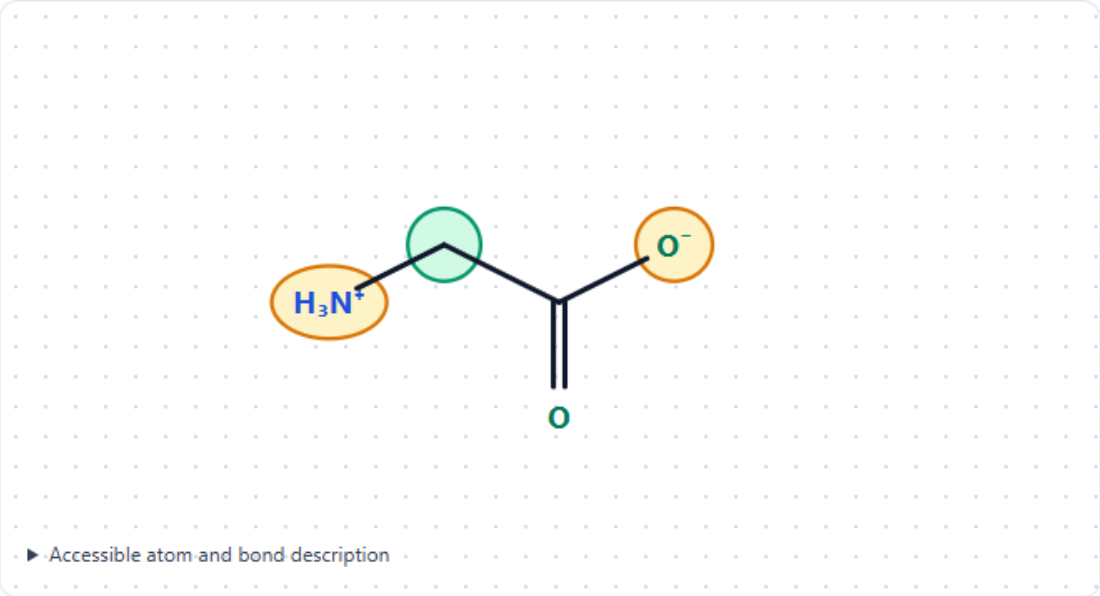
amino-acids:step-2 | 2. Glycine exception

Glycine has two hydrogens at its alpha carbon and is achiral. Its neutral net charge near suitable pH does not mean it lacks charged sites.

Displayed molecule ID: glycine

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Screenshot: amino-acids-step-2



amino-acids:step-3 | 3. Relative configuration

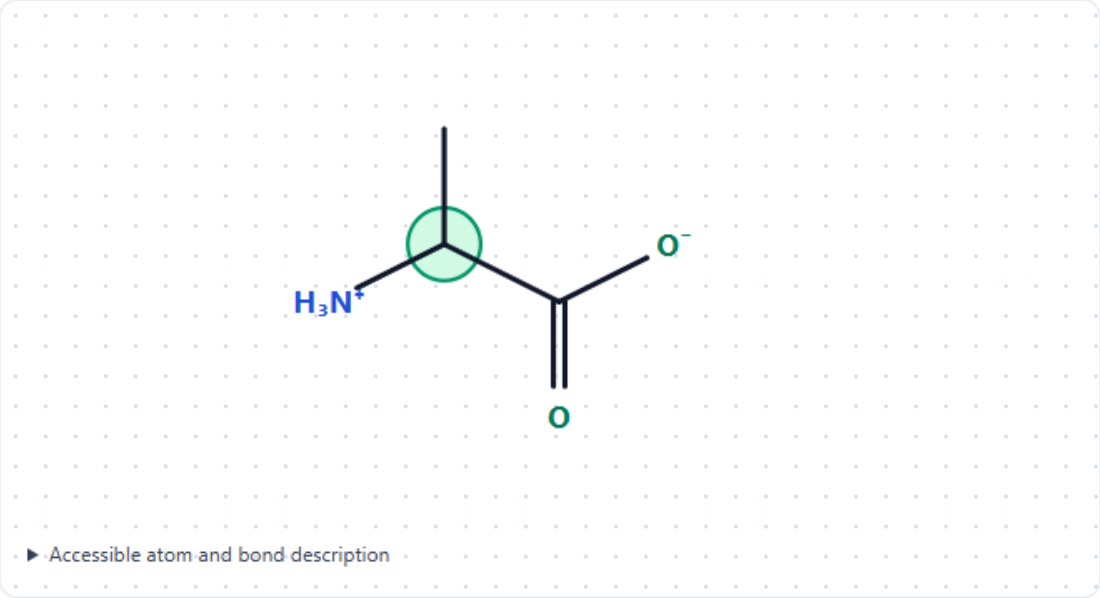
The separate Fischer projection below shows L-alanine: carboxyl above, methyl below, amino left. The skeletal canvas above specifies connectivity without configuration. In the Fischer projection, horizontal bonds point toward you. L does not specify

the optical-rotation sign.

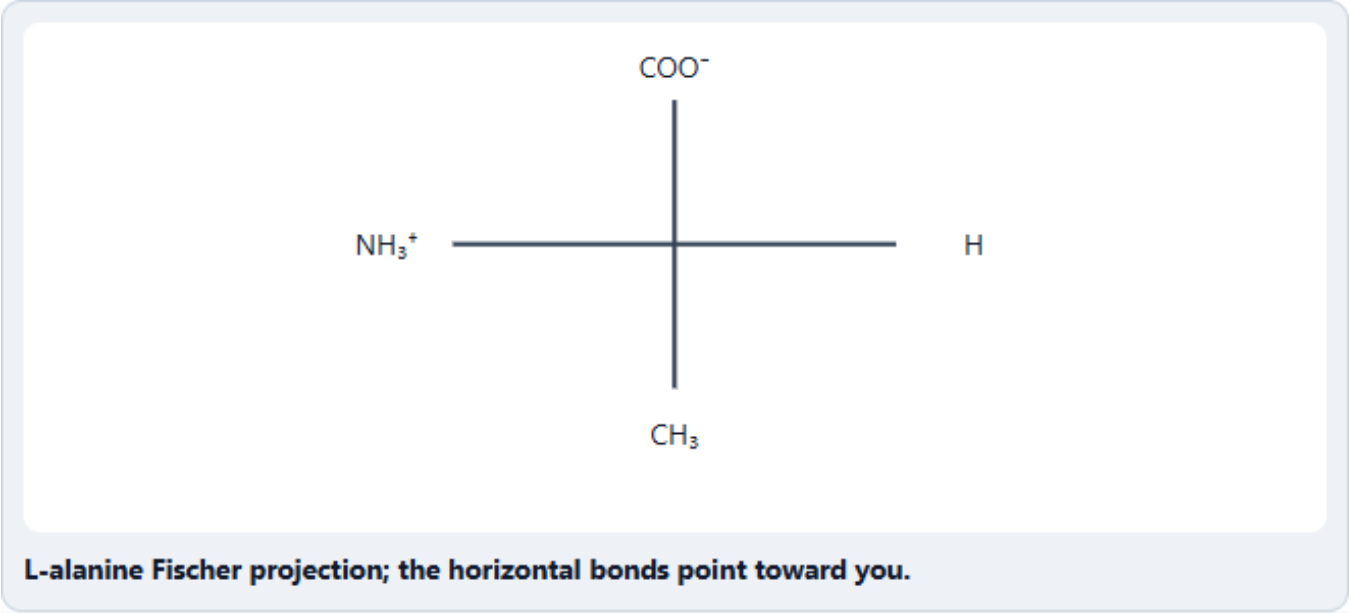
Displayed molecule ID: alanine

{"highlights": {"prin": [1]}, "revealName": false}

Screenshot: amino-acids-step-3



Screenshot: amino-acids-step-3-diagram



{"diagramSpecification": {"type": "fischer", "top": "COO⁻", "bottom": "CH₃", "left": "NH₃⁺", "right": "H", "caption": "L-alanine Fischer projection; the horizontal bonds point toward you."}}

peptides | Peptide bonds and protein structure

An amide links amino-acid residues

peptides:rule | Core Rules

A peptide bond joins one residue's carbonyl carbon to the next residue's nitrogen. Resonance makes the peptide group approximately planar with restricted C–N rotation. Sequence is conventionally read N-terminus → C-terminus. Primary sequence supports secondary structures, tertiary folding, and sometimes quaternary assembly. Net peptide formation is a condensation/dehydration, eliminating water from amino and carboxyl groups in formal bookkeeping. Activation enables synthesis; hydrolysis consumes water to cleave the amide, with enzymes or suitable strong acid/base and often heat. Basic cleavage is base-promoted and gives carboxylate, not necessarily a neutral acid before workup.

Schematic summary: An amide links amino-acid residues

Linkage — Carbonyl C to next residue N

Peptide group — Approximately planar; restricted C–N rotation

Sequence — N-terminus to C-terminus

peptides:cue | Memory Cues

N to C is the reading direction. Peptide C–N is the restricted bond; adjacent backbone single bonds supply much of the flexibility.

Schematic summary: Choose the case that applies

Peptide C–N — Partial double-bond character restricts rotation

Adjacent backbone single bonds — Supply much of the flexibility

peptides:trap | Common Mistakes / MCAT Traps

A peptide bond is not a C–C bond between two alpha carbons. Peptide formation is thermodynamically and kinetically enabled by activation in cells; plain amino acids in water do not rapidly polymerize by themselves.

Schematic summary: Shortcut versus a better check

Avoid — Peptide bonds join two alpha carbons

Check — Carbonyl C to N; formation requires activation

peptides:app | MCAT Applications

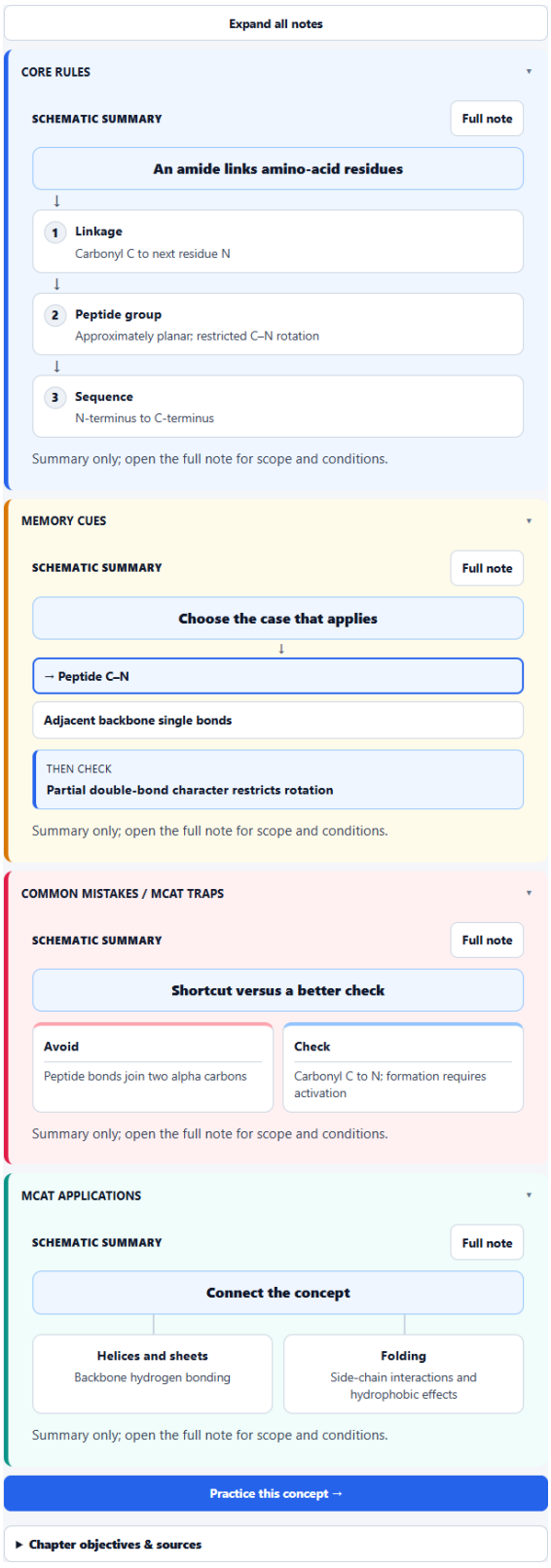
Hydrogen bonding organizes alpha helices and beta sheets; hydrophobic effects and side-chain interactions help stabilize folded proteins.

Schematic summary: Connect the concept

Helices and sheets — Backbone hydrogen bonding

Folding — Side-chain interactions and hydrophobic effects

Screenshot: peptides-visuals



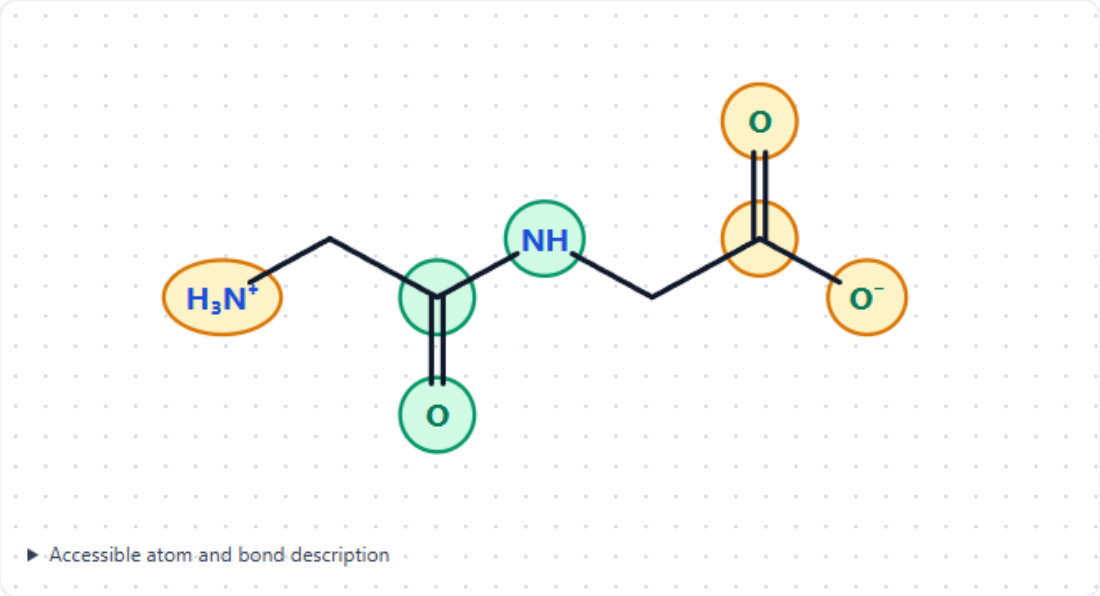
peptides:step-1 | 1. Find the amide linkage

The internal nitrogen is bonded directly to the preceding carbonyl carbon. The terminal amino and carboxyl groups remain available for ionization.

Displayed molecule ID: glycylglycine

{"highlights": {"prin": [2, 3, 4], "sub": [0, 6, 7, 8]}, "revealName": true}

Screenshot: peptides-step-1



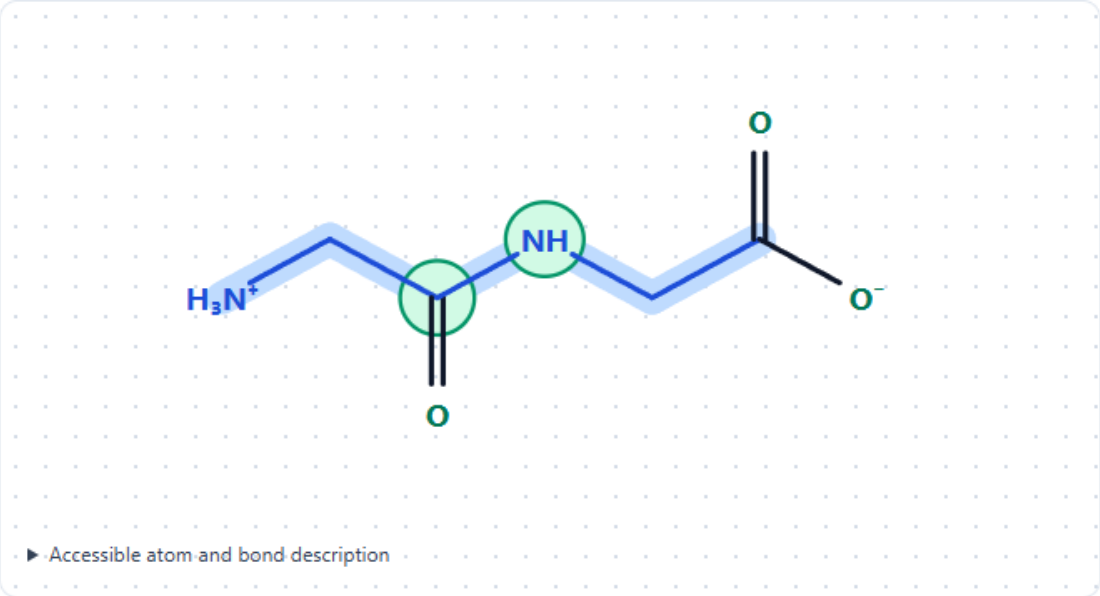
peptides:step-2 | 2. Trace the backbone

Follow N–Cα–C(=O)–N–Cα–C(=O). The peptide C–N link is approximately planar because the N lone pair participates in resonance.

Displayed molecule ID: glycylglycine

{"highlights": {"path": [0, 1, 2, 4, 5, 6], "prin": [2, 4]}, "revealName": false}

Screenshot: peptides-step-2



synthesis | Strecker and Gabriel amino-acid strategies

Track where the carboxyl carbon comes from

synthesis:rule | Core Rules

In the Strecker route, an aldehyde reacts through an imine/iminium stage with ammonia and cyanide to form an alpha-aminonitrile. Nitrile hydrolysis gives the carboxyl group. A Gabriel–malonic ester route installs protected nitrogen and an alkyl side chain, then hydrolyzes and decarboxylates to an alpha amino acid. One Strecker teaching reagent set uses NH_4Cl/KCN, followed by aqueous acid and heat: imine/iminium formation, cyanide carbon addition, then nitrile hydrolysis. These are reaction-recognition details, not an experimental procedure.

Schematic summary: Track where the carboxyl carbon comes from

Strecker stage — Aldehyde, ammonia and cyanide chemistry

Aminonitrile — Follow the nitrile carbon

Hydrolysis — Nitrile carbon becomes carboxyl carbon

synthesis:cue | Memory Cues

Strecker: nitrile carbon becomes carboxyl carbon. Gabriel–malonate: alkylation builds the side chain; decarboxylation removes one carboxyl group. In the Gabriel–malonic ester variant, potassium phthalimide substitutes at the brominated carbon of diethyl bromomalonate. Deprotonation of the protected aminomalonate then enables carbon alkylation with a suitable alkyl halide by SN2. Hydrolysis/deprotection and decarboxylation give the amino acid; the phthalimide framework can leave as phthalic acid under acidic hydrolysis conditions. The electrophile must be compatible with SN2; tertiary halides do not serve as a universal choice.

Schematic summary: Choose the case that applies

Strecker — Cyanide carbon becomes carboxyl carbon

Gabriel–malonate variant — Alkylation builds the side chain; decarboxylation removes one carboxyl group

synthesis:trap | Common Mistakes / MCAT Traps

Achiral starting materials and conditions usually give a racemic chiral amino-acid product; glycine is achiral. Ordinary Gabriel amine synthesis and the malonic-ester amino-acid variant are related but not identical.

Schematic summary: Shortcut versus a better check

Avoid — Every amino-acid product must be racemic

Check — Glycine is achiral; substrate and chiral conditions matter

synthesis:app | MCAT Applications

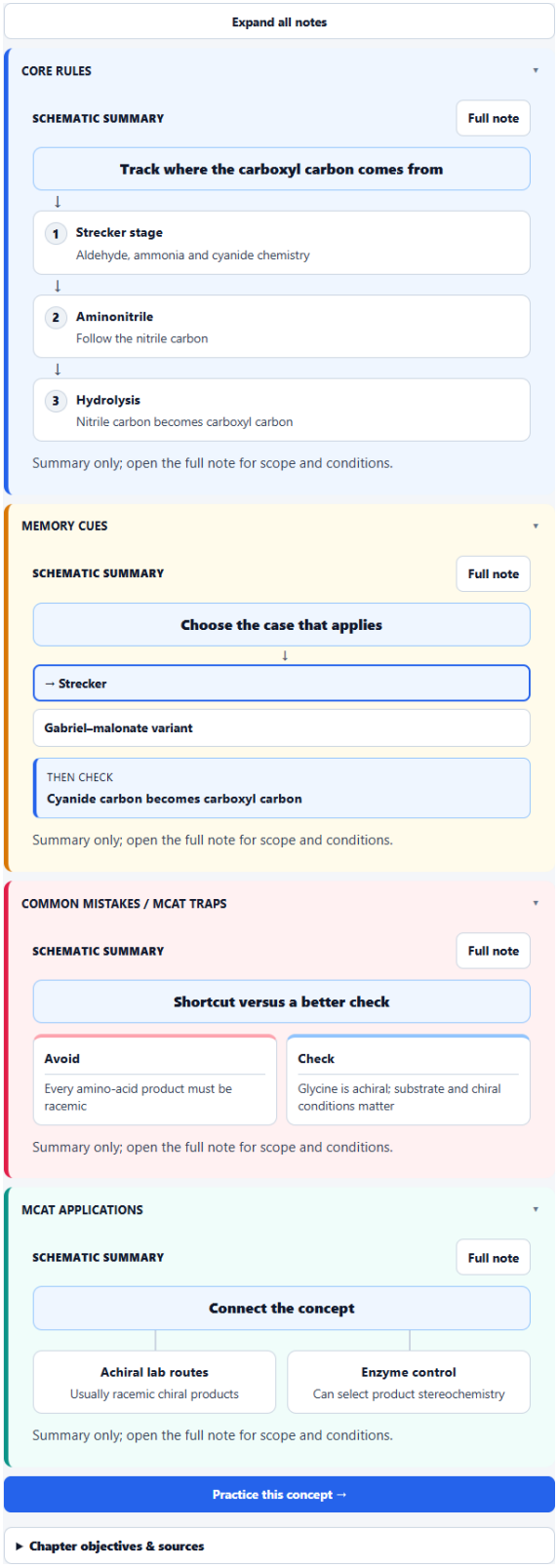
Laboratory racemic synthesis contrasts with enzyme-controlled production of stereochemically selected amino acids.

Schematic summary: Connect the concept

Achiral lab routes — Usually racemic chiral products

Enzyme control — Can select product stereochemistry

Screenshot: synthesis-visuals



synthesis:step-1 | 1. Aminonitrile intermediate

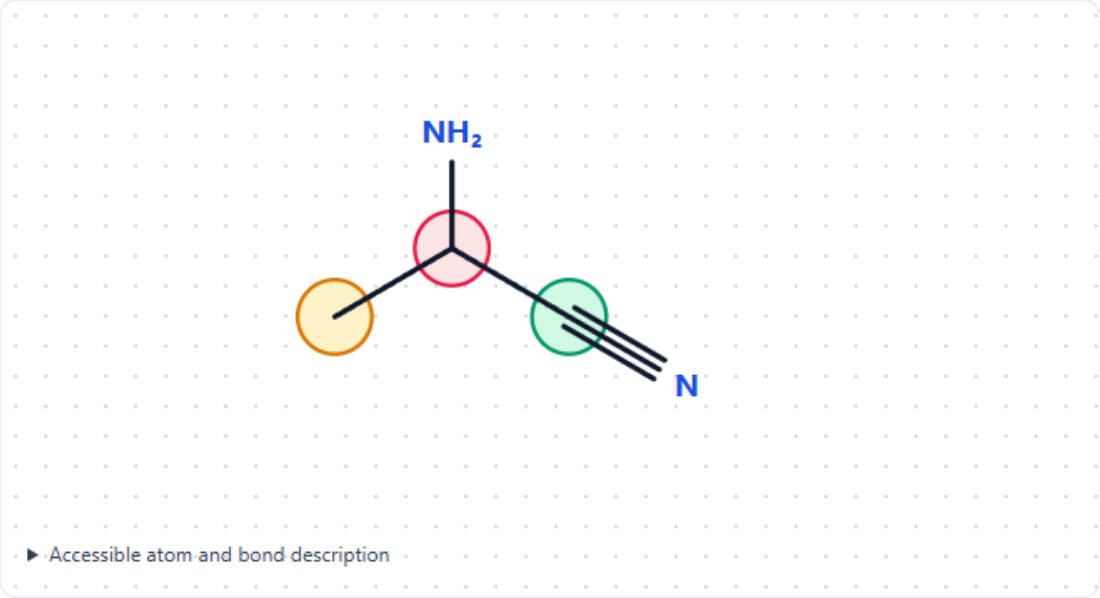
The central carbon carries an amino group, methyl, H, and a nitrile carbon. This is the carbon skeleton leading to alanine on nitrile hydrolysis. Green tracks the nitrile-derived future carboxyl carbon, red the future alpha carbon, and orange the

retained methyl side chain. The nitrile N is not retained in the amino acid.

Displayed molecule ID: aminonitrile

{"highlights": {"prin": [2], "sub": [0], "alpha": [1]}, "revealName": true}

Screenshot: synthesis-step-1



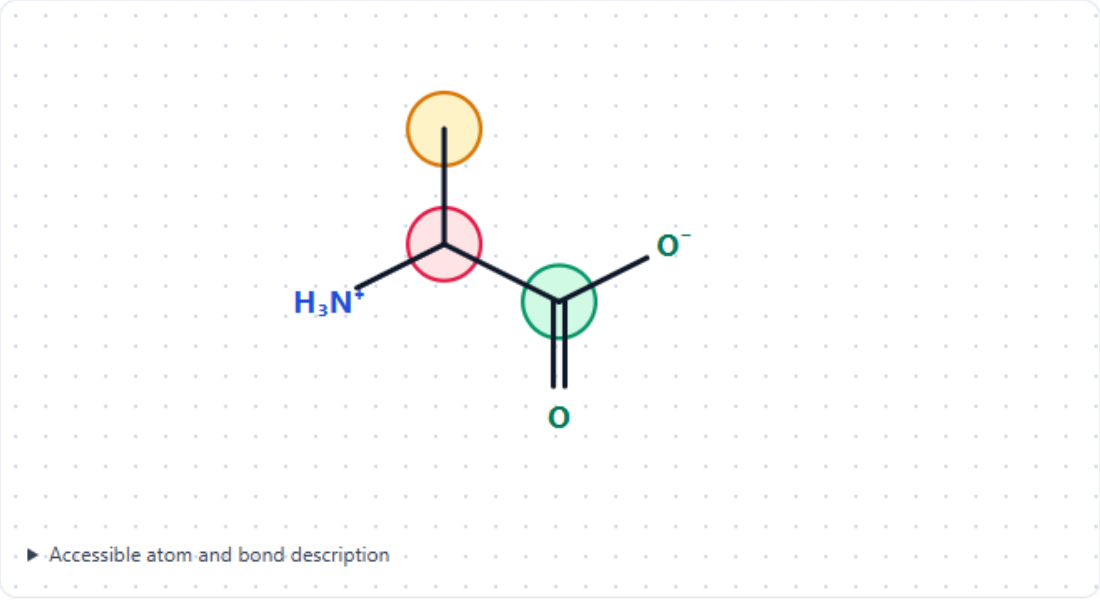
synthesis:step-2 | 2. Carbon bookkeeping

The nitrile carbon becomes the amino acid's carboxyl carbon. The original aldehyde carbon becomes its alpha carbon; the methyl side chain is retained. Protonation state then depends on pH.

Displayed molecule ID: alanine

{"highlights": {"prin": [2], "sub": [5], "alpha": [1]}, "revealName": true}

Screenshot: synthesis-step-2



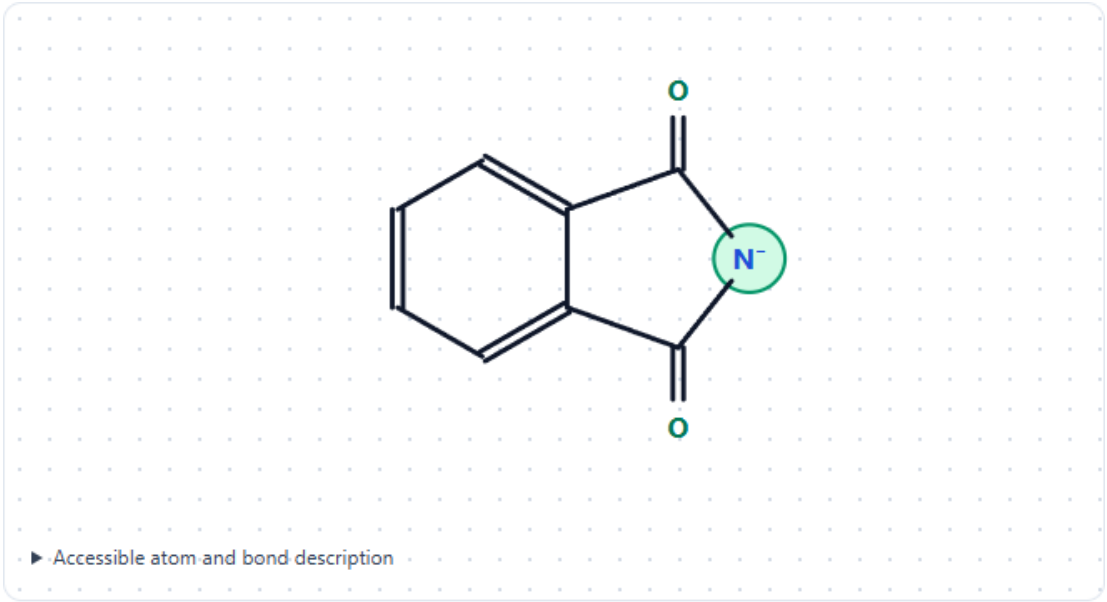
synthesis:step-3 | 3. Alternative protected route

The phthalimide anion shown supplies protected nitrogen; its potassium counterion is omitted. N substitution on diethyl bromomalonate installs the protected amino group. Subsequent malonate carbon alkylation installs a side chain; hydrolysis/deprotection and decarboxylation then give an alpha amino acid. This reagent canvas is not the complete reaction sequence.

Displayed molecule ID: phthalimide-anion

{"highlights": {"prin": [7]}, "revealName": true}

Screenshot: synthesis-step-3



phosphorus | Phosphates and phosphoryl transfer

Phosphoesters differ from phosphoanhydrides

phosphorus:rule | Core Rules

A phosphate ester contains P–O–C; a phosphodiester has two such ester links. A phosphoanhydride contains P–O–P. Phosphoric acid has successive dissociations, so charge depends on pH. Formal P=O Lewis drawings are bookkeeping models of delocalized bonding. Organic phosphate esters attach carbon groups through P–O–C. Inorganic orthophosphate (Pi) lacks such ester links and includes protonation states set by pH. Pyrophosphate (PPi) joins two phosphate units through P–O–P; its charge also depends on pH.

Schematic summary: Phosphoesters differ from phosphoanhydrides

P–O–C — Phosphate ester linkage

Two ester links — Phosphodiester context

P–O–P — Phosphoanhydride linkage

phosphorus:cue | Memory Cues

DNA/RNA backbone: phosphodiester. Between ATP phosphate groups: phosphoanhydride. Approximate aqueous phosphoric-acid pKa values are 2.1, 7.2 and 12.3; $H_2PO_4^-$/HPO_4^{2-} buffers near neutral pH. Temperature and ionic conditions shift the values. Useful buffering regions lie near each pKa, often within about one pH unit; capacity depends on concentration and component ratio. Three separated regions do not give uniformly effective buffering at all intervening pH values.

Schematic summary: Choose the case that applies

DNA/RNA backbone — Phosphodiester

Between ATP phosphates — Phosphoanhydride

Near-neutral phosphate buffer — $H_2PO_4^-$ / HPO_4^{2-}; distinct pKa regions

phosphorus:trap | Common Mistakes / MCAT Traps

Breaking a bond requires energy. Favorable ATP hydrolysis reflects the overall free-energy difference between reactants and solvated products, including resonance, electrostatics, and solvation; energy is not released simply by breaking one bond.

Schematic summary: Shortcut versus a better check

Avoid — Breaking one ATP bond releases energy

Check — Overall free-energy difference, including solvation and electrostatics

phosphorus:app | MCAT Applications

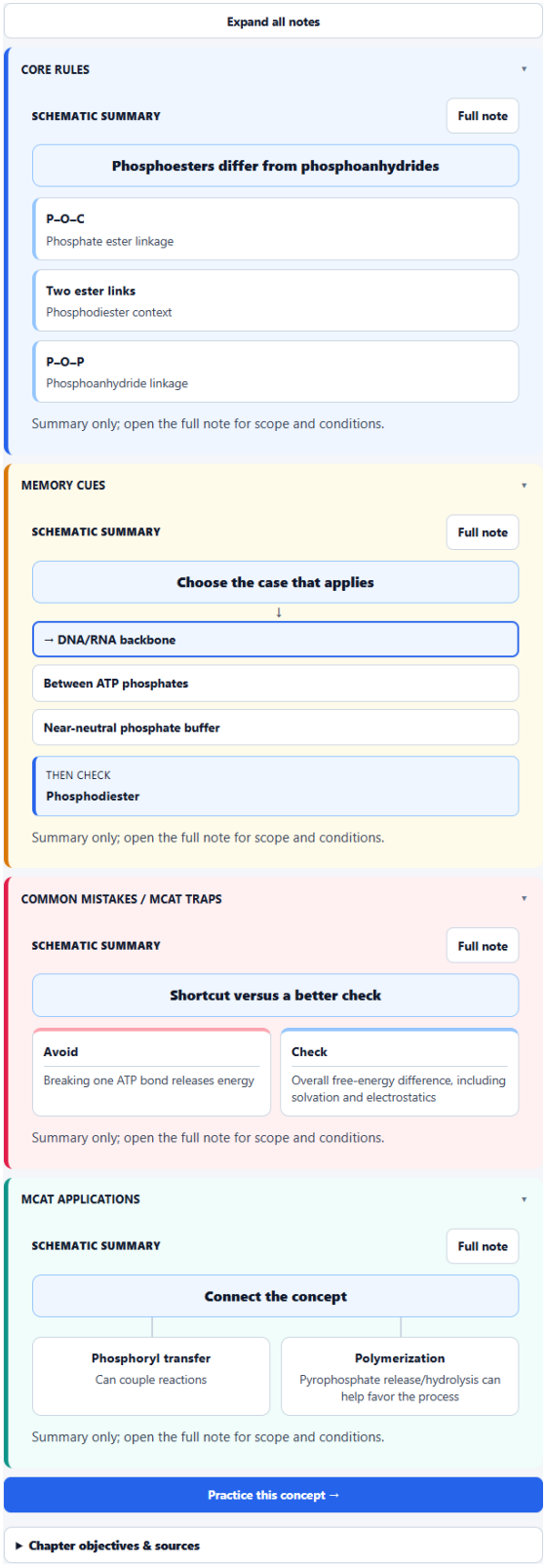
Phosphoryl transfer can couple reactions. Nucleic-acid synthesis releases pyrophosphate, whose subsequent hydrolysis can help favor polymerization.

Schematic summary: Connect the concept

Phosphoryl transfer — Can couple reactions

Polymerization — Pyrophosphate release/hydrolysis can help favor the process

Screenshot: phosphorus-visuals



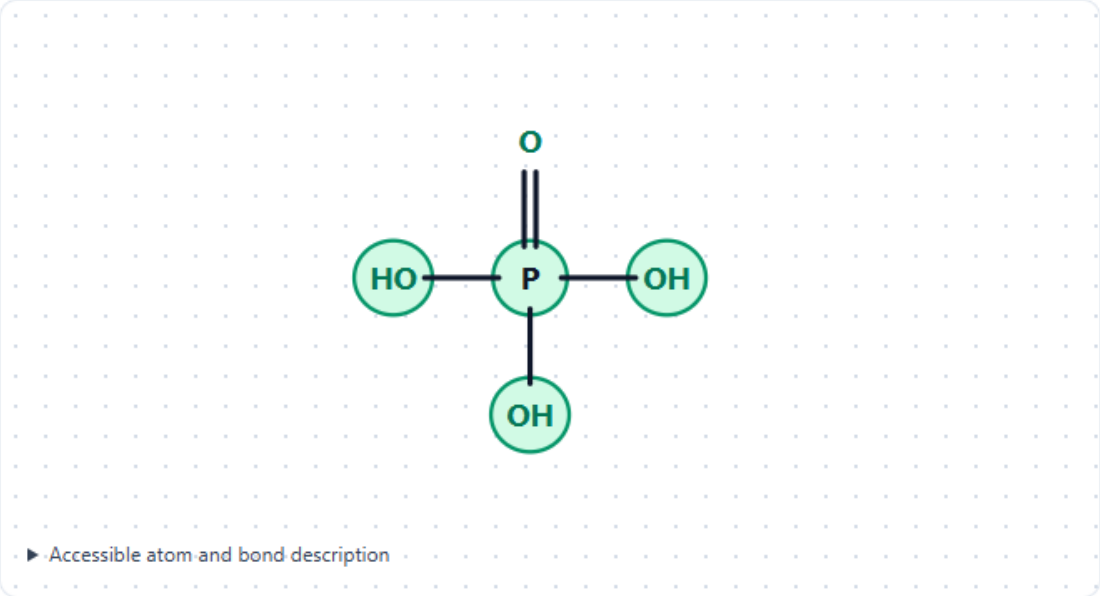
phosphorus:step-1 | 1. Acidic phosphate

The neutral phosphoric-acid representation has three OH groups. Sequential deprotonation gives different phosphate ions; this drawing is not the predominant form at every pH.

Displayed molecule ID: phosphoricacid

{"highlights": {"prin": [0, 2, 3, 4]}, "revealName": true}

Screenshot: phosphorus-step-1



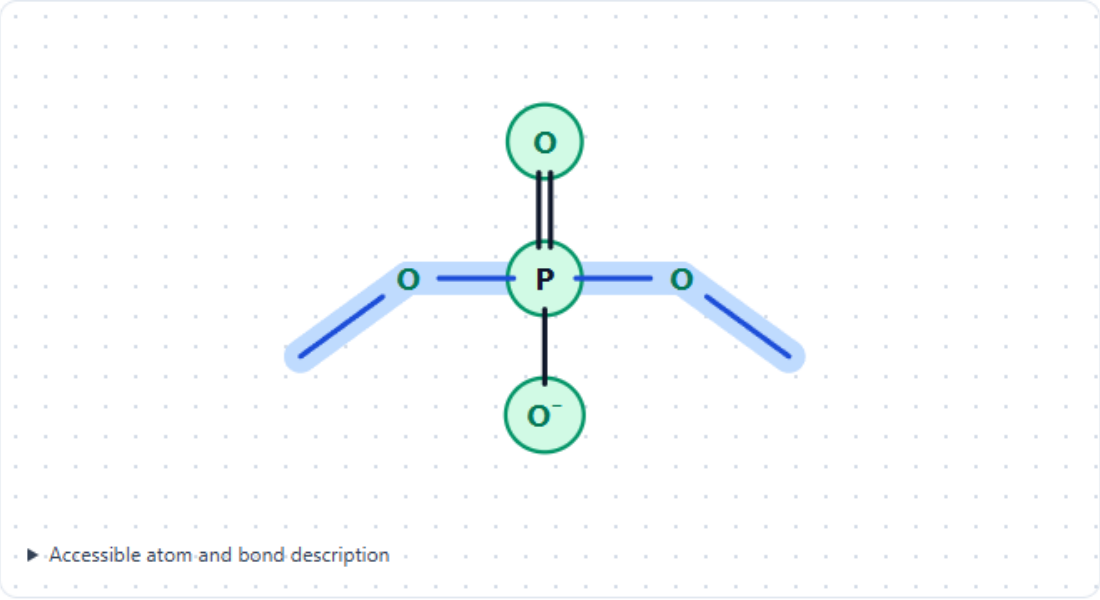
phosphorus:step-2 | 2. Phosphodiester model

Dimethyl phosphate illustrates two P–O–C connections and a negative charge. The nucleic-acid backbone replaces the methyl groups with sugar-derived groups.

Displayed molecule ID: dimethylphosphate

{"highlights": {"path": [5, 2, 0, 3, 6], "prin": [0, 1, 4]}, "revealName": true}

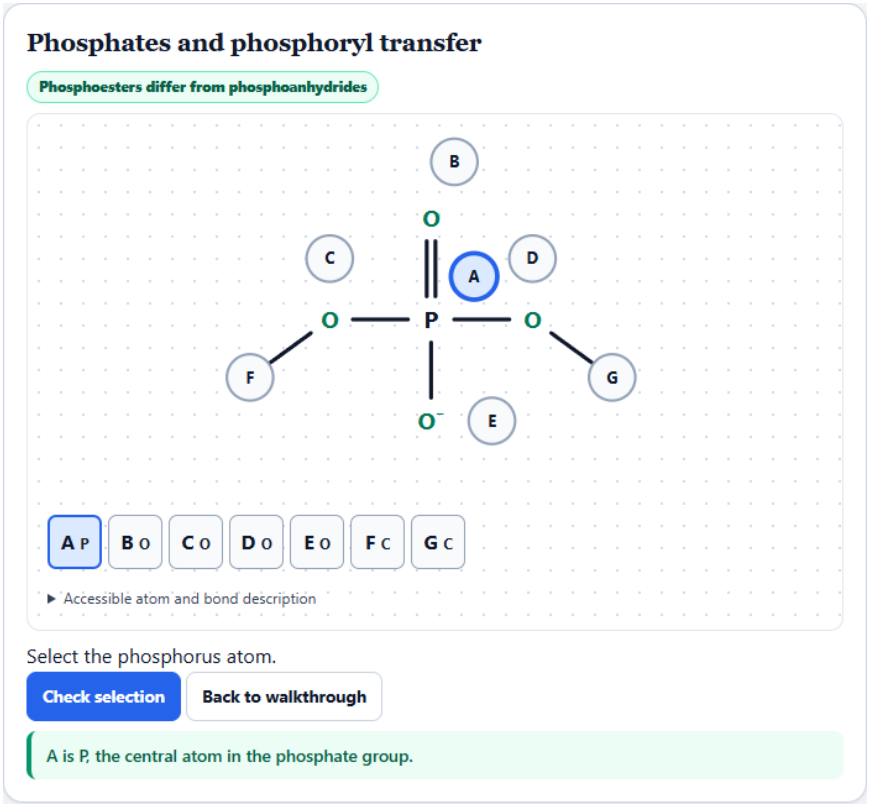
Screenshot: phosphorus-step-2



Embedded challenge: Select the phosphorus atom.

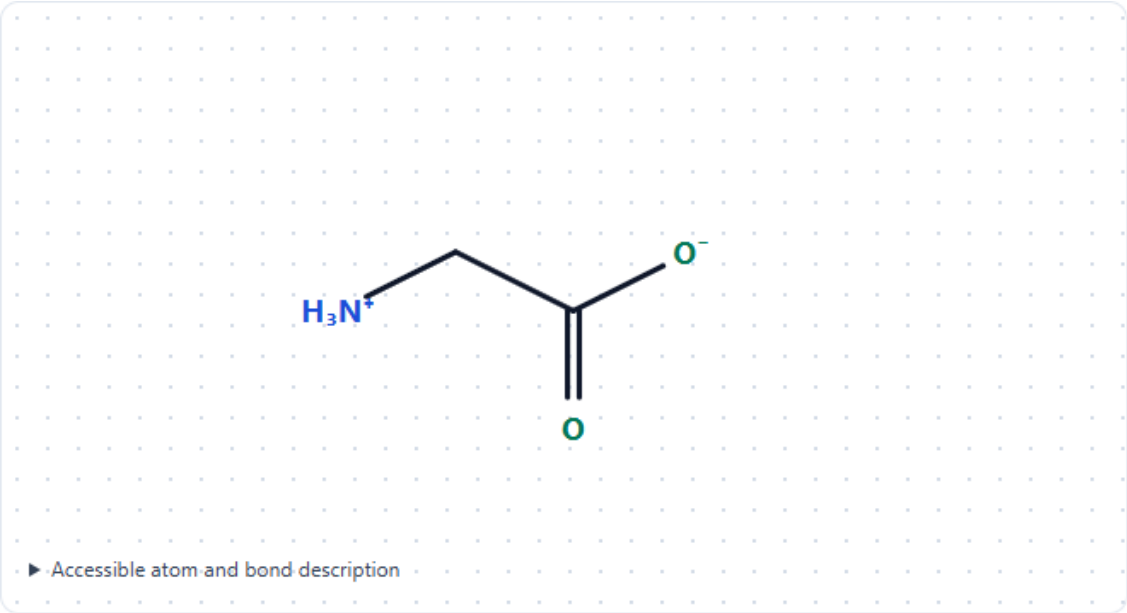
{"prompt": "Select the phosphorus atom.", "answer": [0], "explanation": "A is P, the central atom in the phosphate group."}

Screenshot: phosphorus-step-2-challenge



glycine-charge | Charge and isoelectric point

Screenshot: glycine-charge-stage-1-molecule



glycine-charge:stage-1 | What is the net formal charge of the displayed zwitterion?

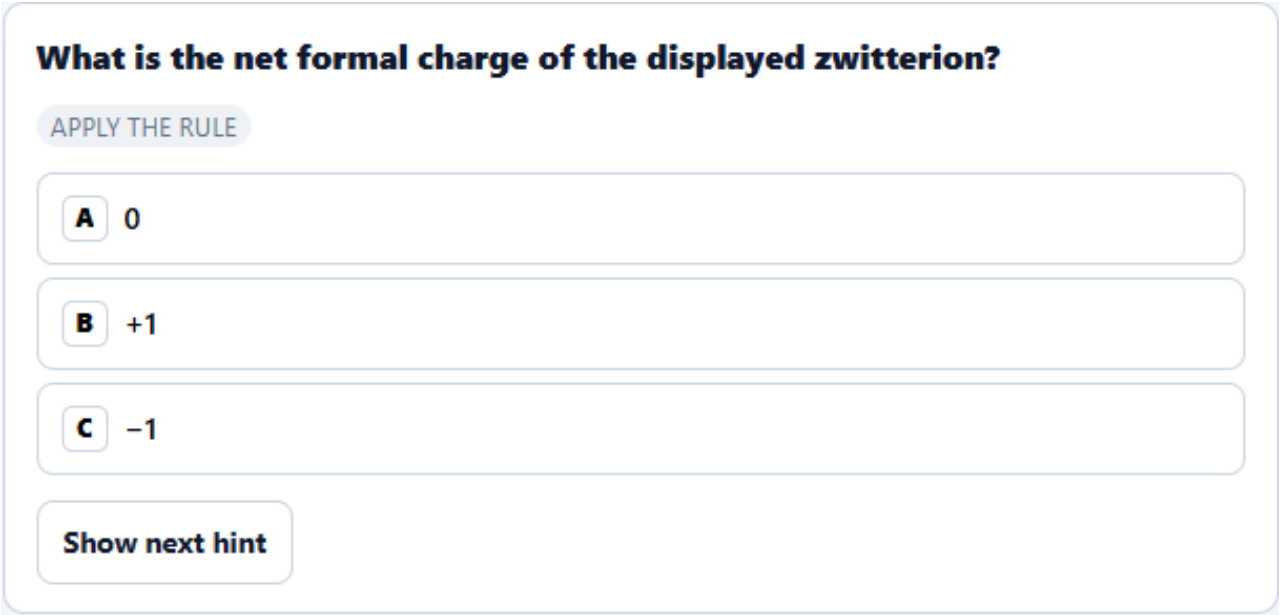
Answer type: choice

Option 0 (A): 0

Option 1 (B): +1

Option 2 (C): −1

Screenshot: glycine-charge-stage-1-question



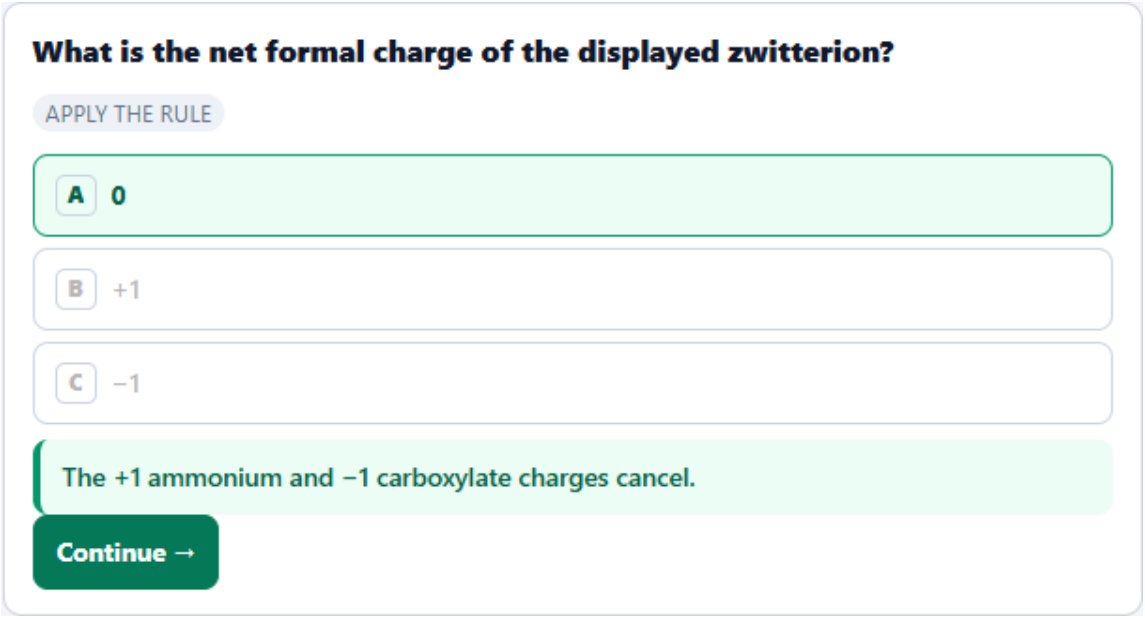
{"authorAnswer": 0, "tolerance": null, "highlightsAfterCorrectAnswer": {}}

Hint 1: Add the labeled formal charges.

Hint 2: Net zero does not mean no charged sites.

Explanation: The +1 ammonium and −1 carboxylate charges cancel.

Screenshot: glycine-charge-stage-1-feedback



glycine-charge:stage-2 | Why is glycine achiral?

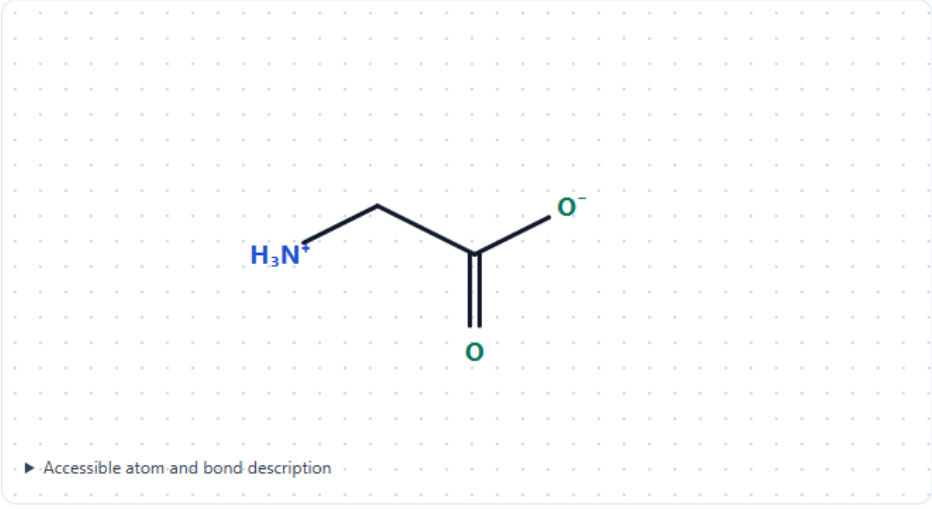
Answer type: choice

Option 0 (A): Its alpha carbon has two H substituents

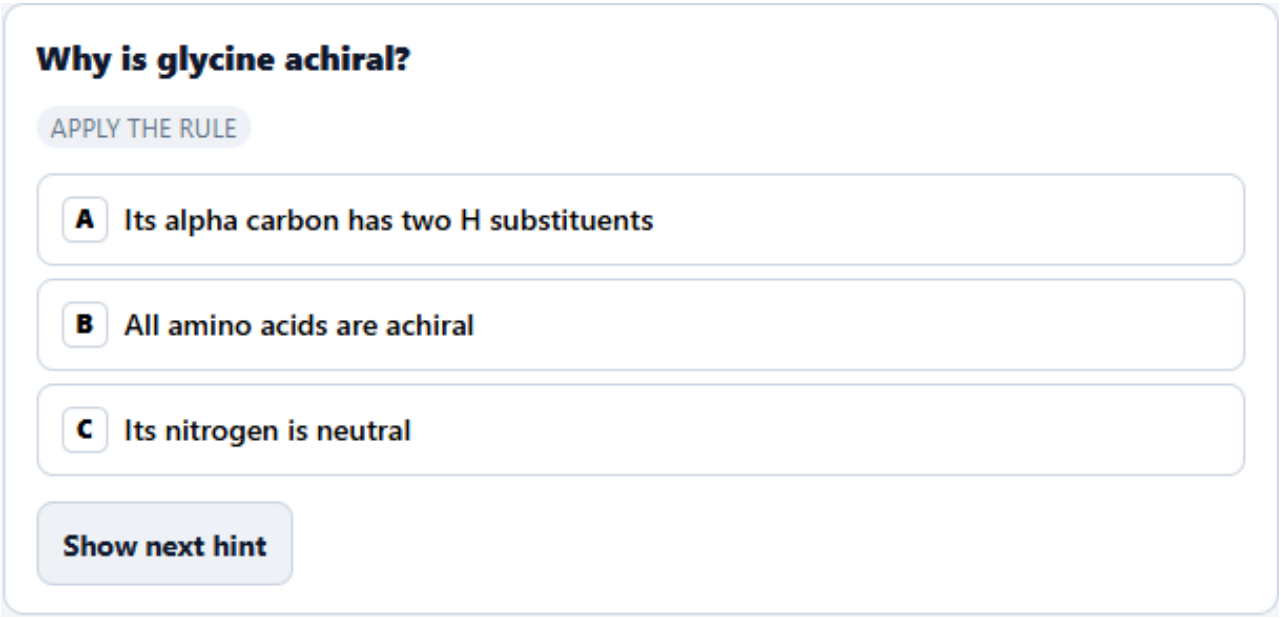
Option 1 (B): All amino acids are achiral

Option 2 (C): Its nitrogen is neutral

Screenshot: glycine-charge-stage-2-molecule



Screenshot: glycine-charge-stage-2-question



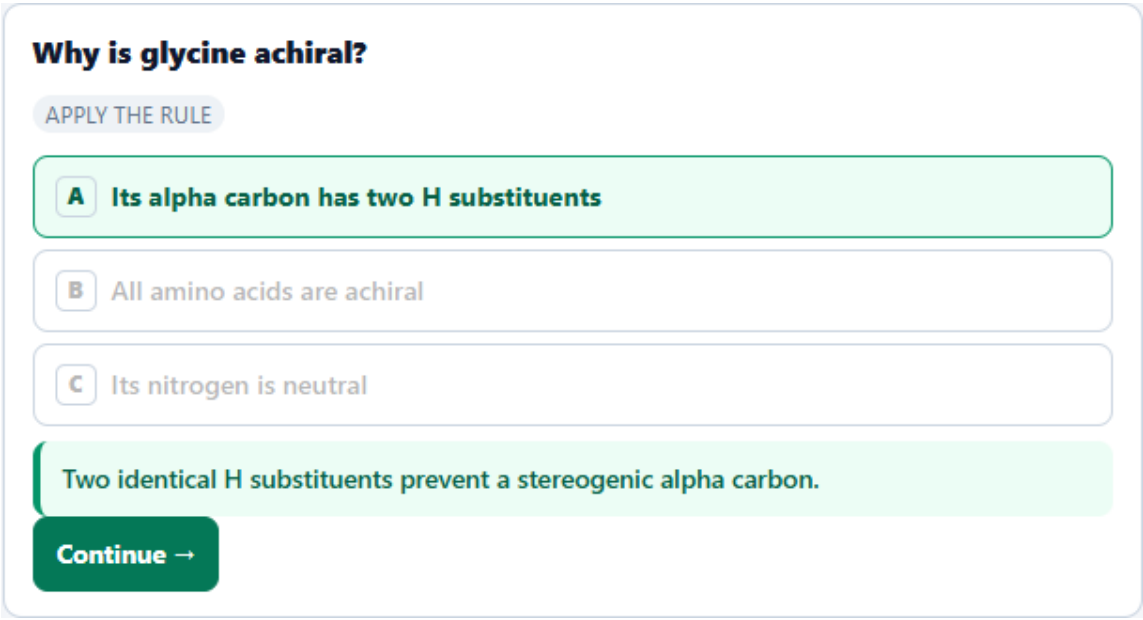
{"authorAnswer": 0, "tolerance": null, "highlightsAfterCorrectAnswer": {}}

Hint 1: Inspect the implicit hydrogens.

Hint 2: The side chain itself is H.

Explanation: Two identical H substituents prevent a stereogenic alpha carbon.

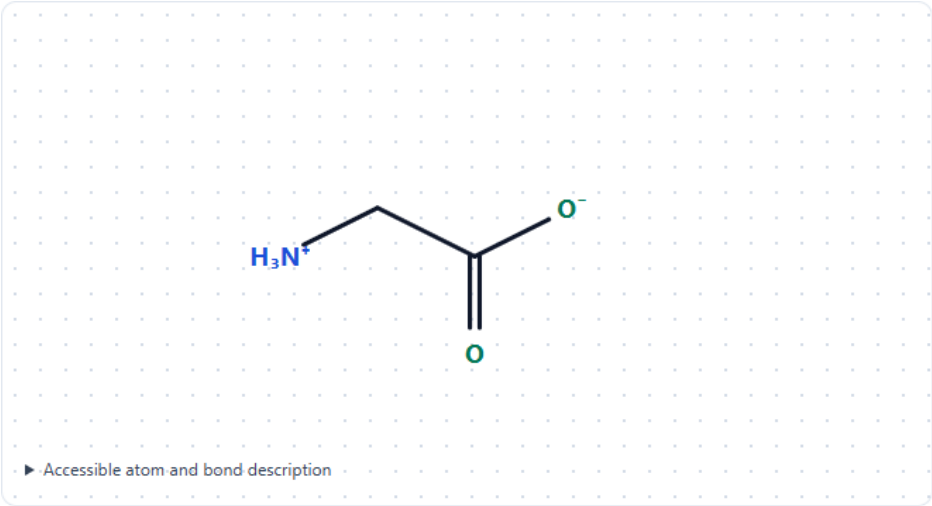
Screenshot: glycine-charge-stage-2-feedback



glycine-charge:stage-3 | Using model pKa values 2.30 and 9.60, calculate pI. Report two decimal places.

Answer type: number | unit: pH units

Screenshot: glycine-charge-stage-3-molecule



Screenshot: glycine-charge-stage-3-question

Using model pKa values 2.30 and 9.60, calculate pI. Report two decimal places.

CALCULATE AND INTERPRET

Numeric answer (pH units)

Check answer

Show next hint

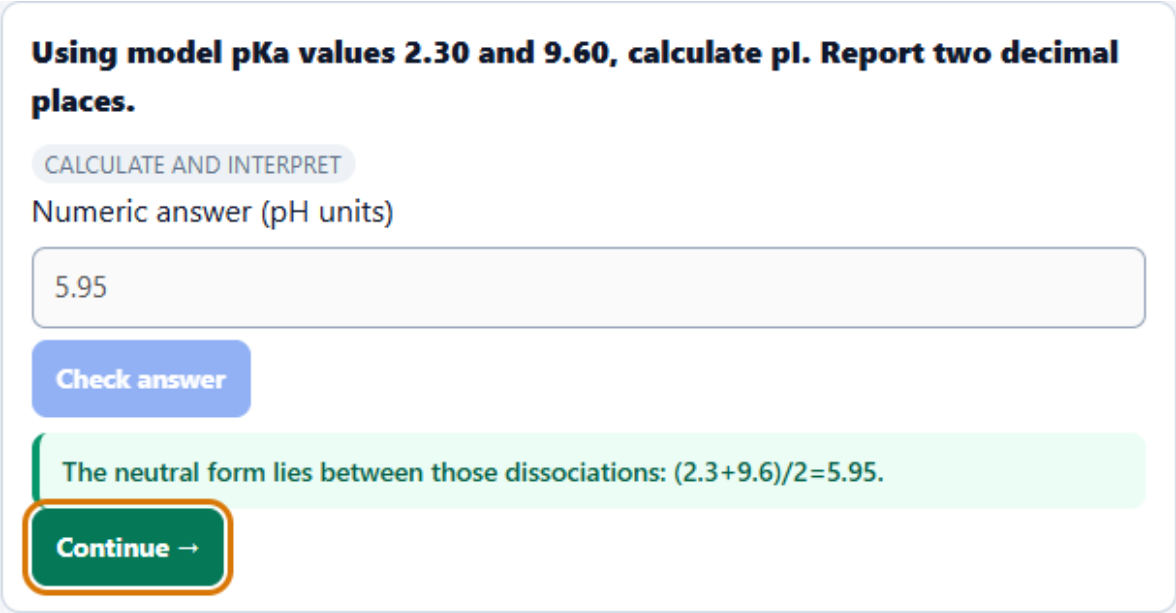
{"authorAnswer": 5.95, "tolerance": 0.005, "highlightsAfterCorrectAnswer": {}}

Hint 1: Use the pKa pair bracketing net zero.

Hint 2: Average the two supplied numbers; retain two decimal places.

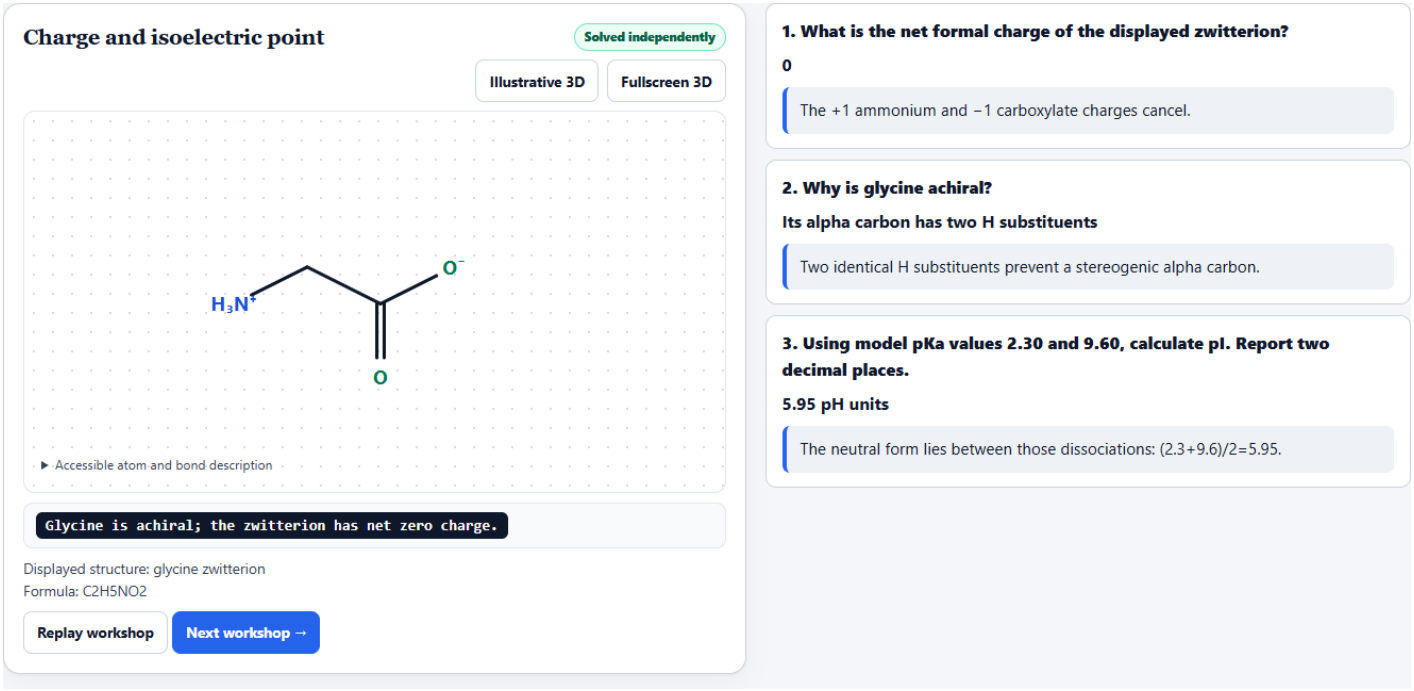
Explanation: The neutral form lies between those dissociations: (2.3+9.6)/2=5.95.

Screenshot: glycine-charge-stage-3-feedback



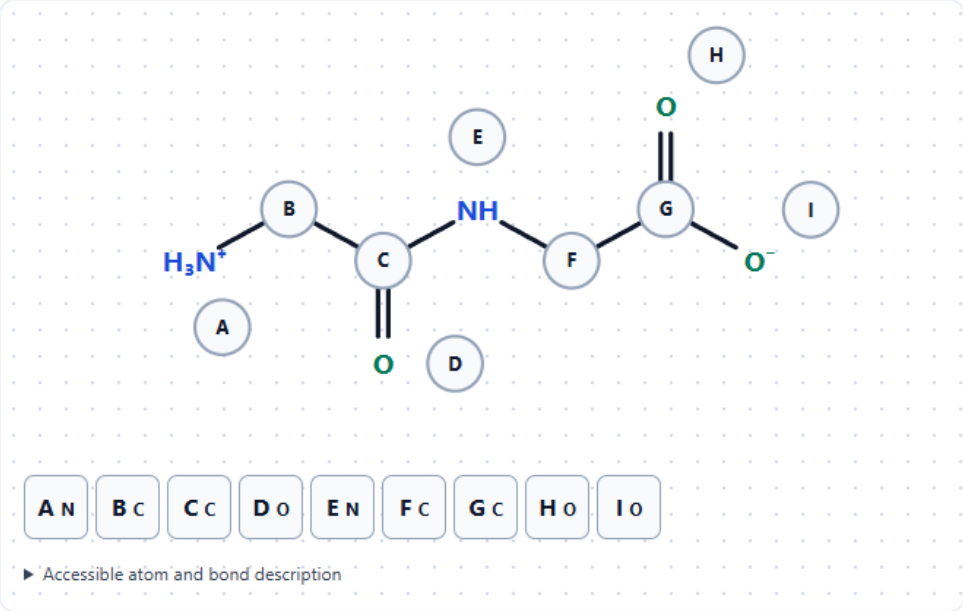
Completion summary: Glycine is achiral; the zwitterion has net zero charge.

Screenshot: glycine-charge-complete



peptide-link | Read a peptide backbone

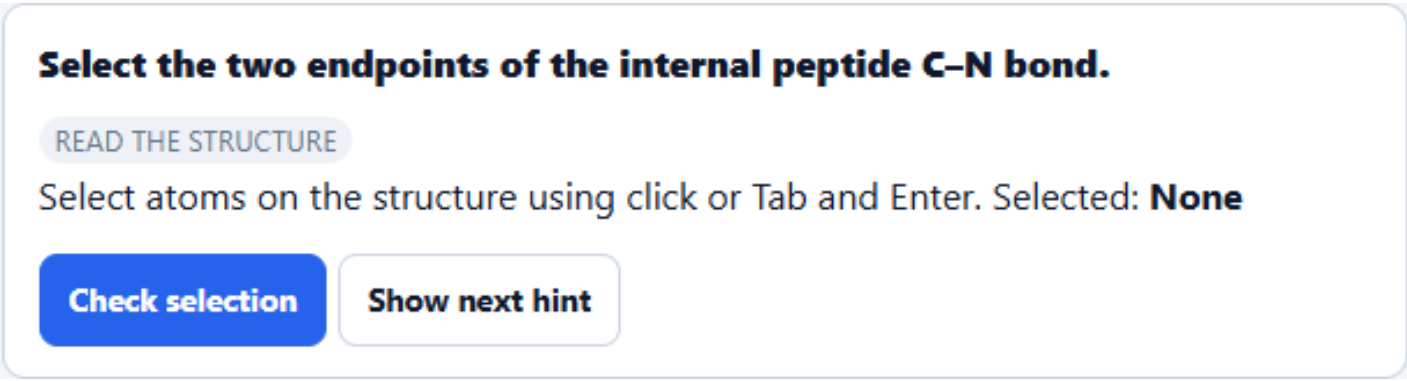
Screenshot: peptide-link-stage-1-molecule



peptide-link:stage-1 | Select the two endpoints of the internal peptide C–N bond.

Answer type: atoms

Screenshot: peptide-link-stage-1-question



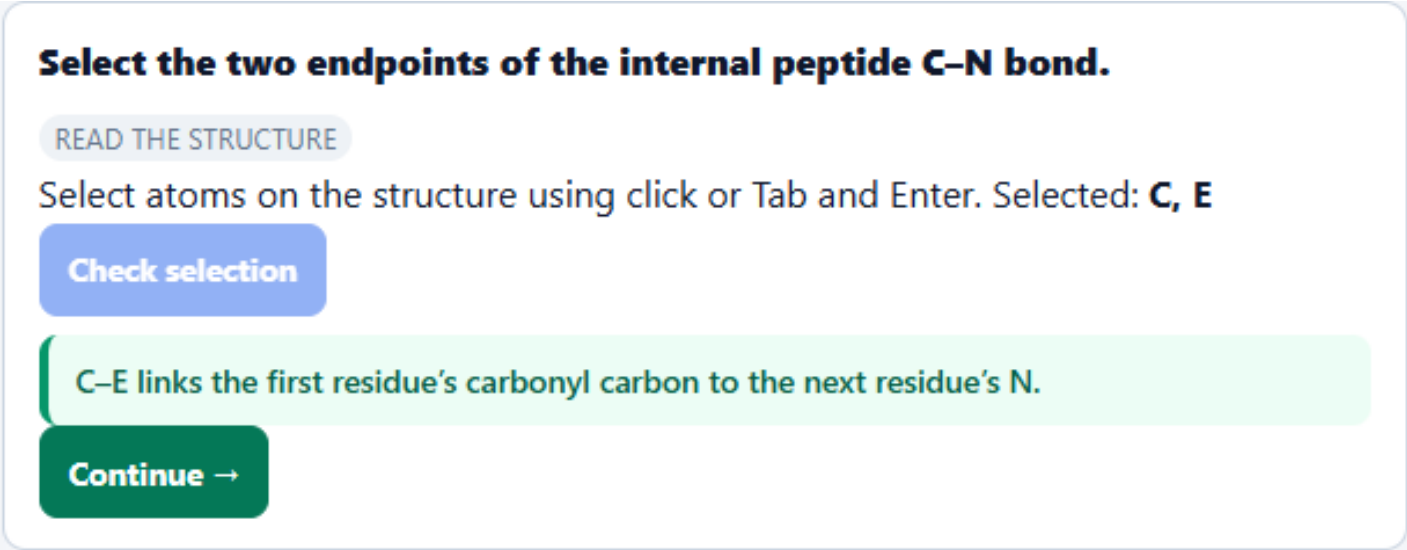
{"authorAnswer": [2, 4], "tolerance": null, "highlightsAfterCorrectAnswer": {"prin": [2, 4]}}

Hint 1: Find C(=O)–N inside the molecule.

Hint 2: Exclude the terminal ammonium nitrogen.

Explanation: C–E links the first residue’s carbonyl carbon to the next residue’s N.

Screenshot: peptide-link-stage-1-feedback



peptide-link:stage-2 | Why is rotation around this bond restricted?

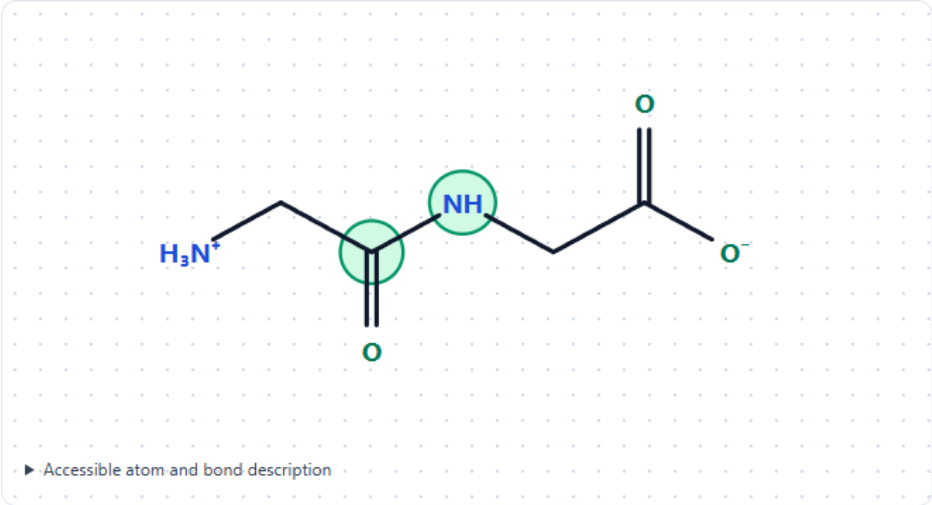
Answer type: choice

Option 0 (A): Amide resonance gives partial double-bond character

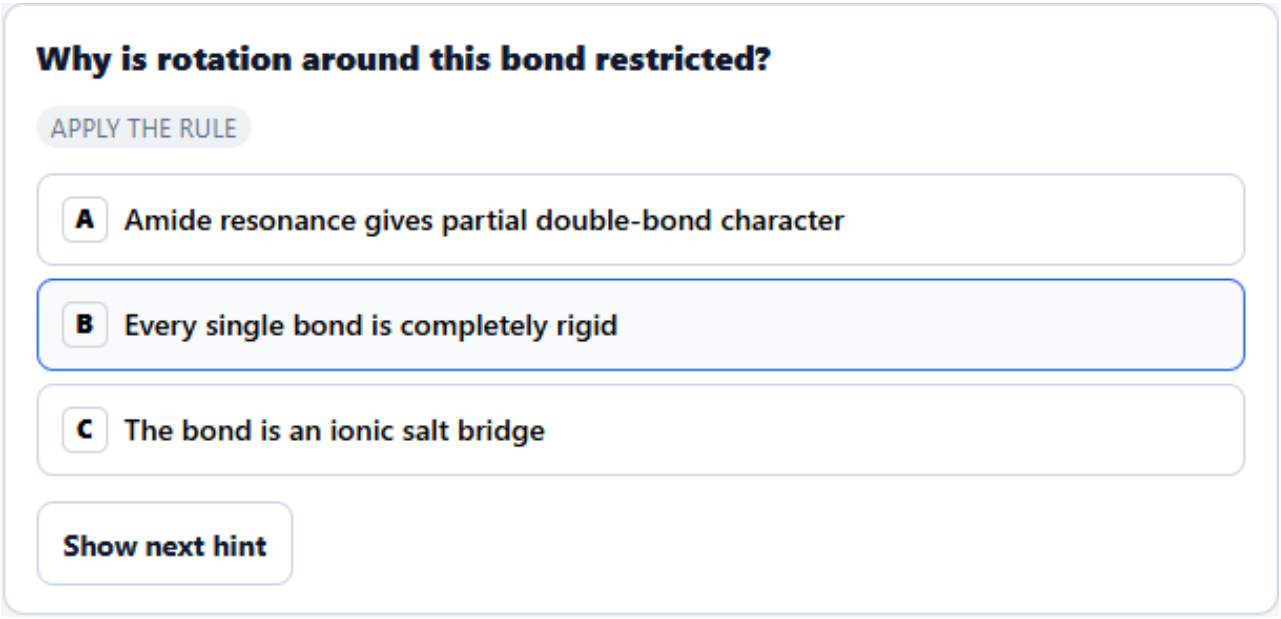
Option 1 (B): Every single bond is completely rigid

Option 2 (C): The bond is an ionic salt bridge

Screenshot: peptide-link-stage-2-molecule



Screenshot: peptide-link-stage-2-question



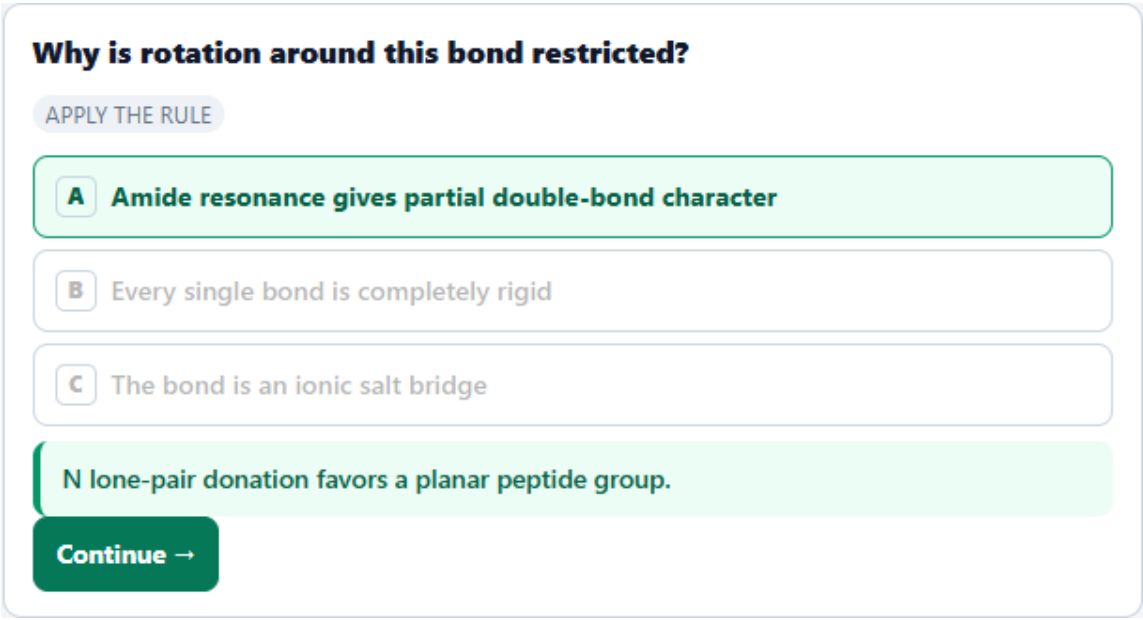
{"authorAnswer": 0, "tolerance": null, "highlightsAfterCorrectAnswer": {}}

Hint 1: Apply the amide resonance model.

Hint 2: Aligned p orbitals stabilize the group.

Explanation: N lone-pair donation favors a planar peptide group.

Screenshot: peptide-link-stage-2-feedback



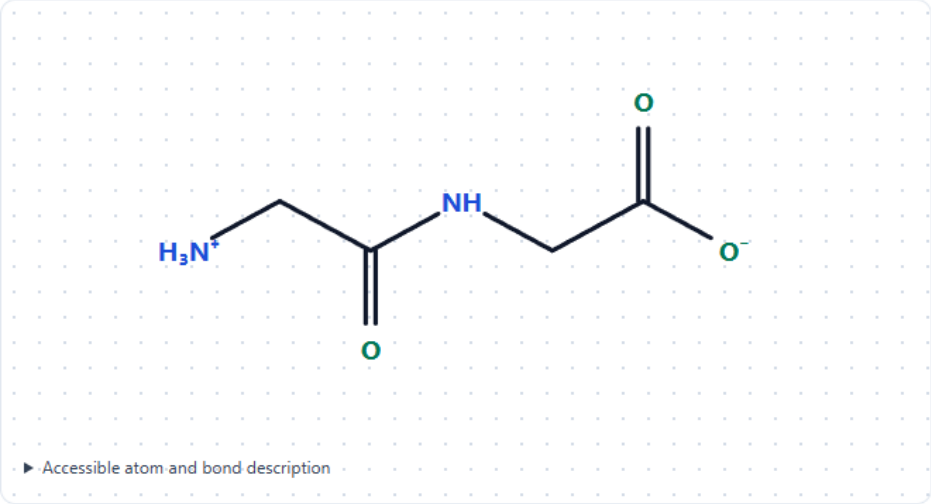
peptide-link:stage-3 | Peptide sequences are read from...

Answer type: choice

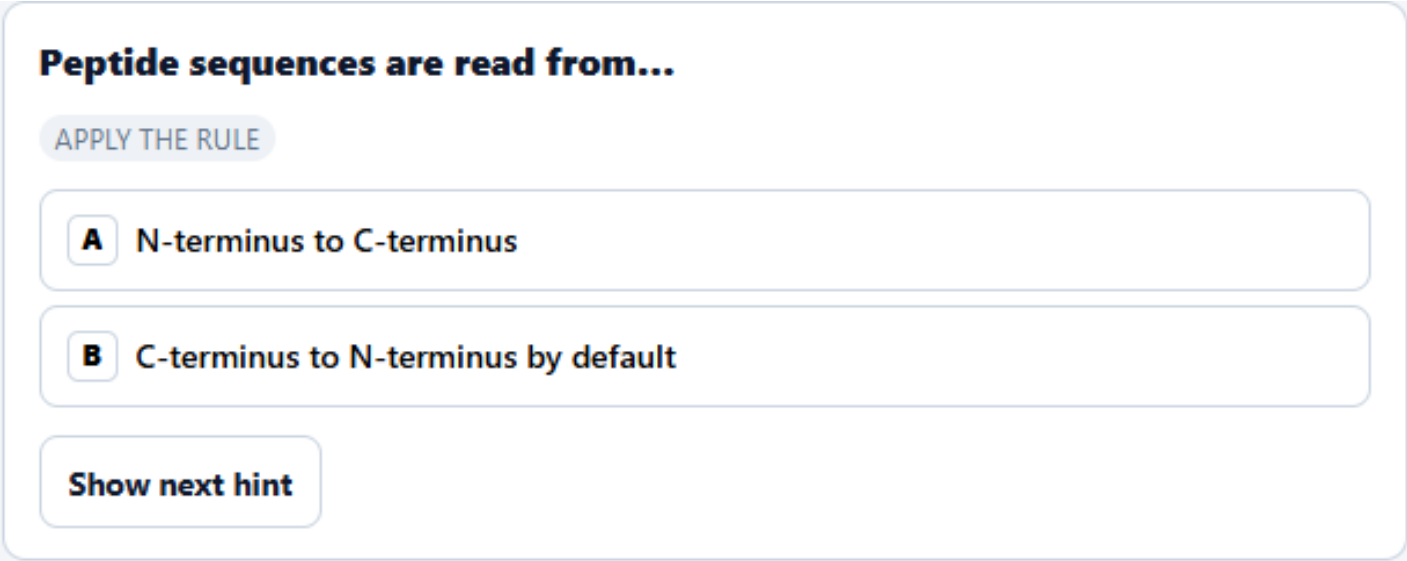
Option 0 (A): N-terminus to C-terminus

Option 1 (B): C-terminus to N-terminus by default

Screenshot: peptide-link-stage-3-molecule



Screenshot: peptide-link-stage-3-question



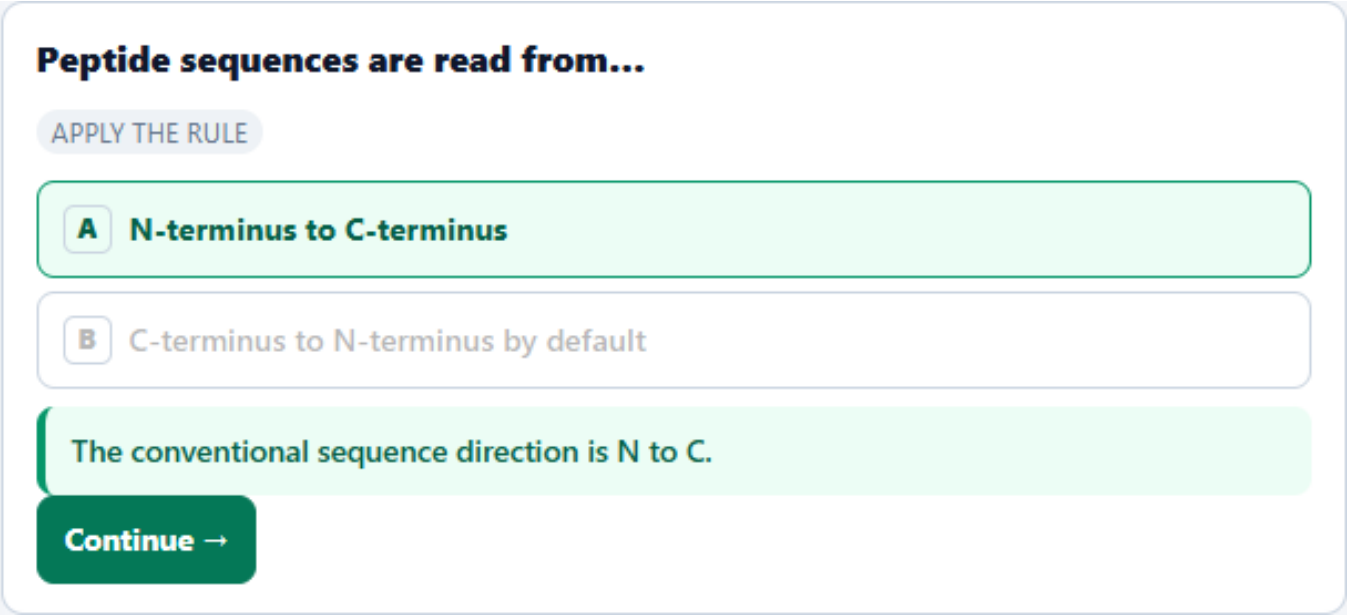
{"authorAnswer": 0, "tolerance": null, "highlightsAfterCorrectAnswer": {}}

Hint 1: Find the free terminal amino group.

Hint 2: Start there and follow the backbone.

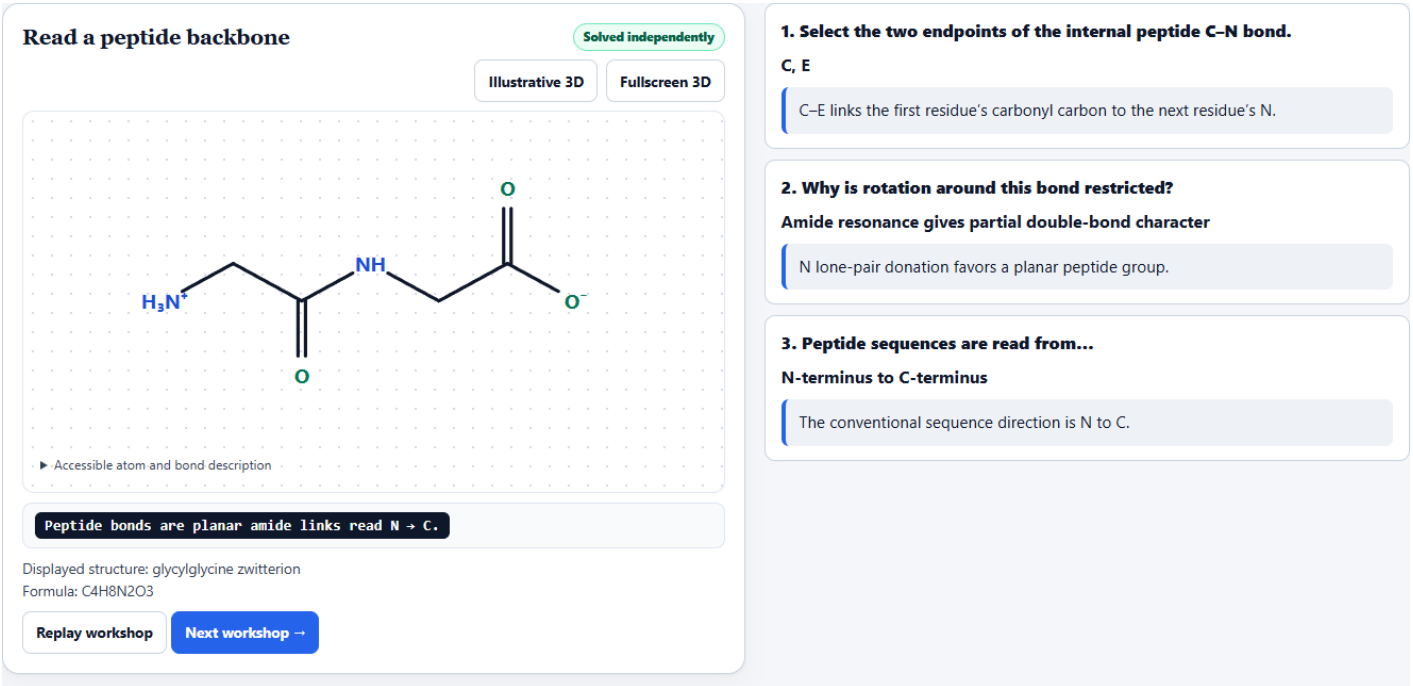
Explanation: The conventional sequence direction is N to C.

Screenshot: peptide-link-stage-3-feedback



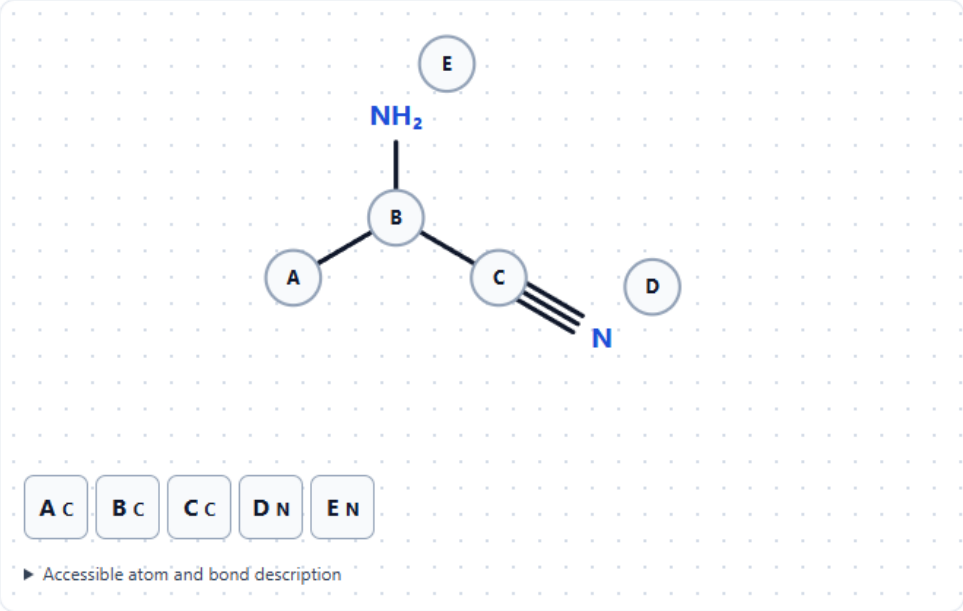
Completion summary: Peptide bonds are planar amide links read N → C.

Screenshot: peptide-link-complete



strecker-map | Track a Strecker intermediate

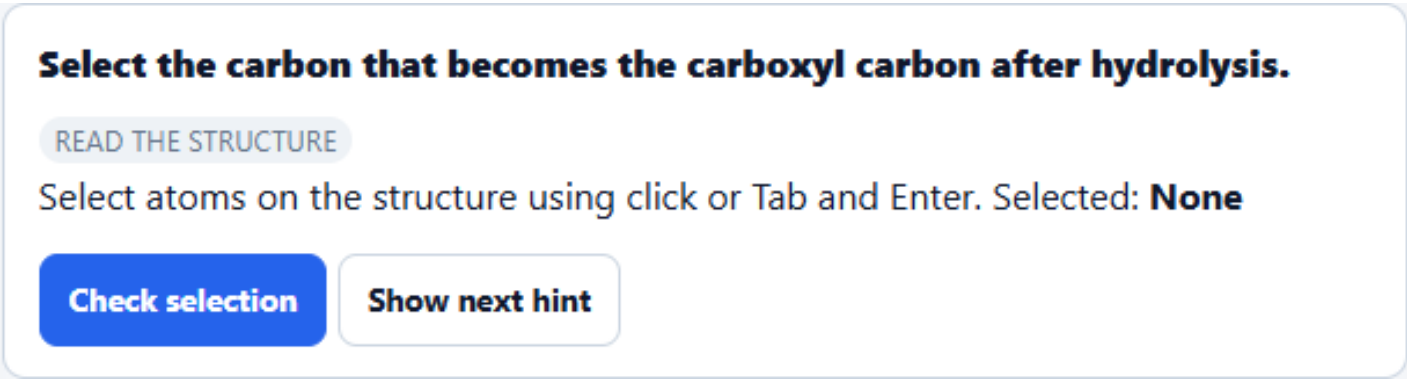
Screenshot: strecker-map-stage-1-molecule



strecker-map:stage-1 | Select the carbon that becomes the carboxyl carbon after hydrolysis.

Answer type: atoms

Screenshot: strecker-map-stage-1-question



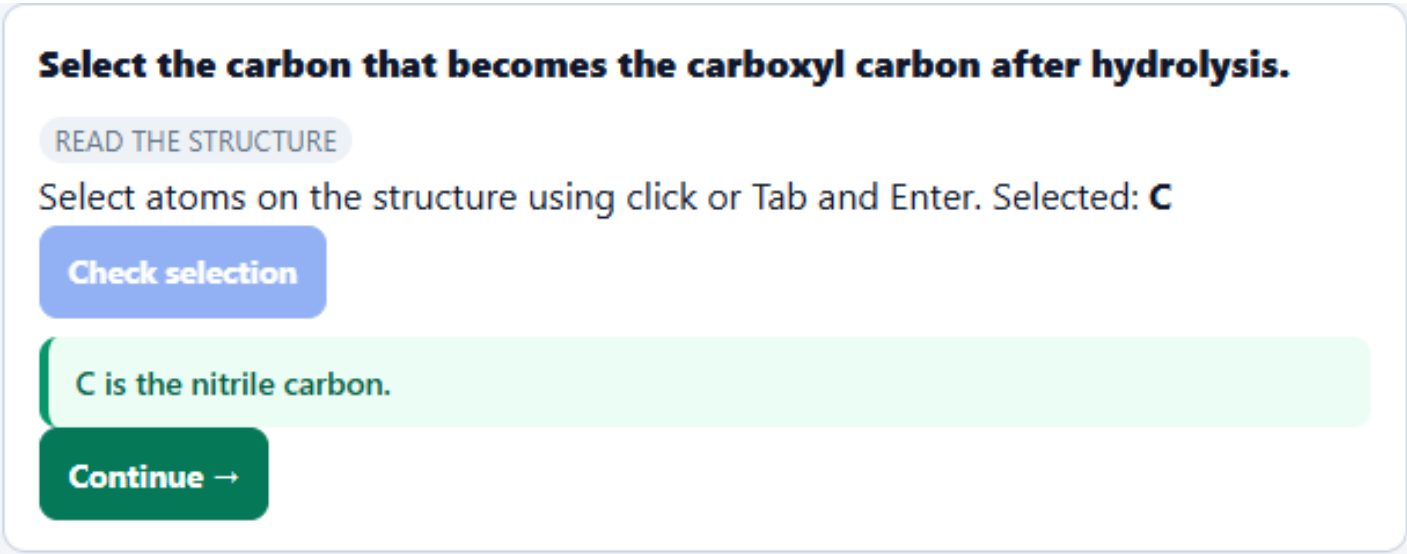
{"authorAnswer": [2], "tolerance": null, "highlightsAfterCorrectAnswer": {"prin": [2]}}

Hint 1: Hydrolysis transforms C≡N into a carboxyl group.

Hint 2: Track the carbon, not the nitrile nitrogen.

Explanation: C is the nitrile carbon.

Screenshot: strecker-map-stage-1-feedback



strecker-map:stage-2 | Which group is retained as the alanine side chain?

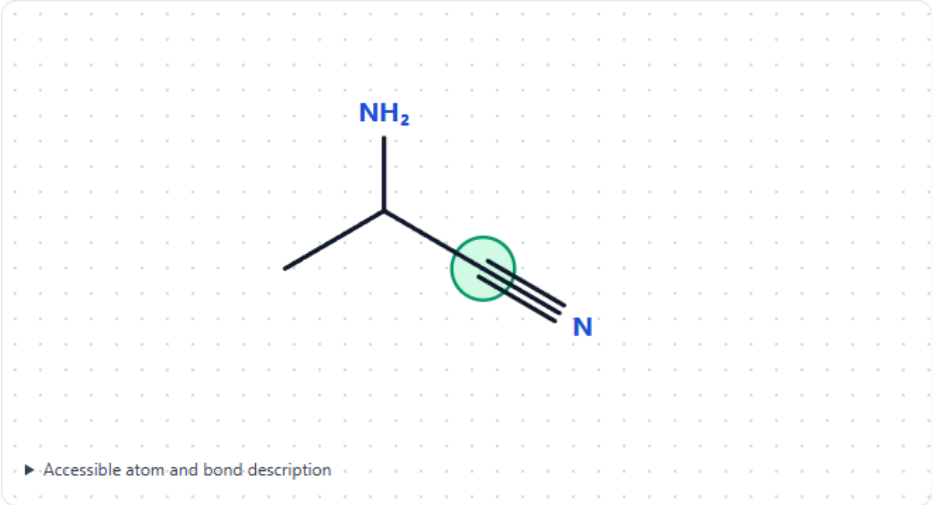
Answer type: choice

Option 0 (A): Methyl

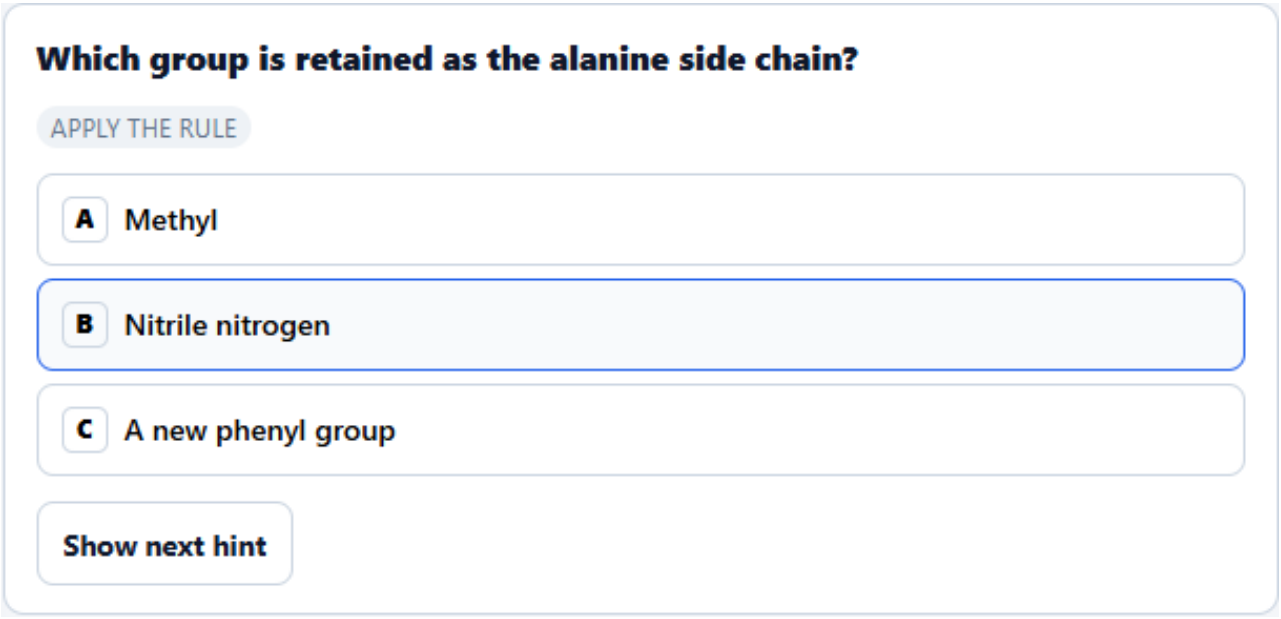
Option 1 (B): Nitrile nitrogen

Option 2 (C): A new phenyl group

Screenshot: strecker-map-stage-2-molecule



Screenshot: strecker-map-stage-2-question



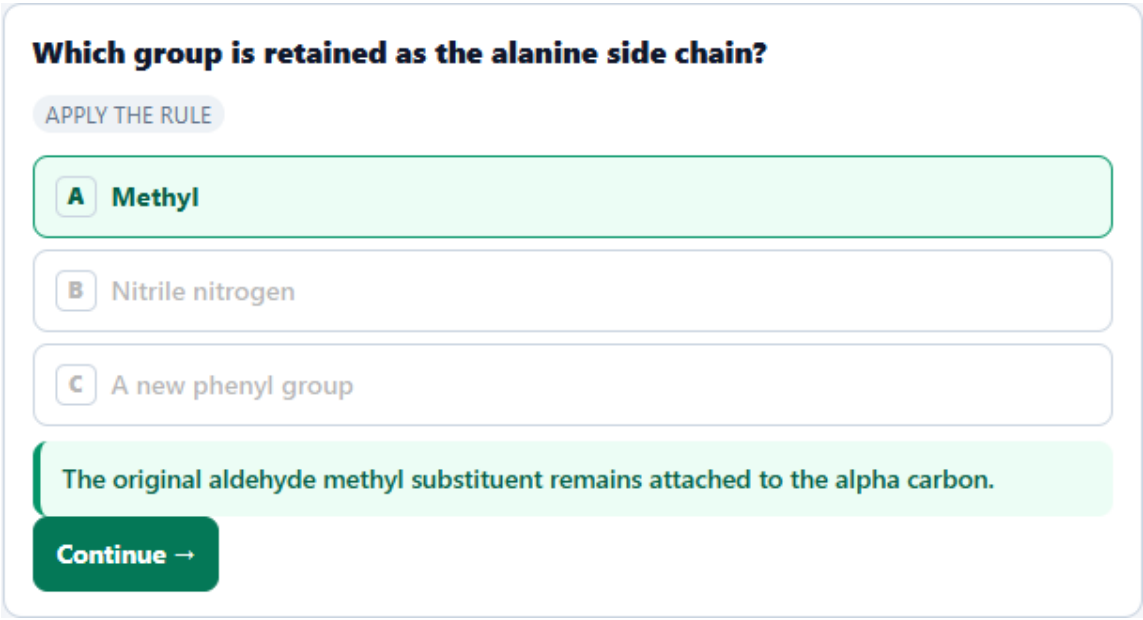
{"authorAnswer": 0, "tolerance": null, "highlightsAfterCorrectAnswer": {"sub": [0]}}

Hint 1: Follow the carbon skeleton.

Hint 2: No new aromatic group is introduced.

Explanation: The original aldehyde methyl substituent remains attached to the alpha carbon.

Screenshot: strecker-map-stage-2-feedback



strecker-map:stage-3 | Under achiral conditions, the chiral product is generally...

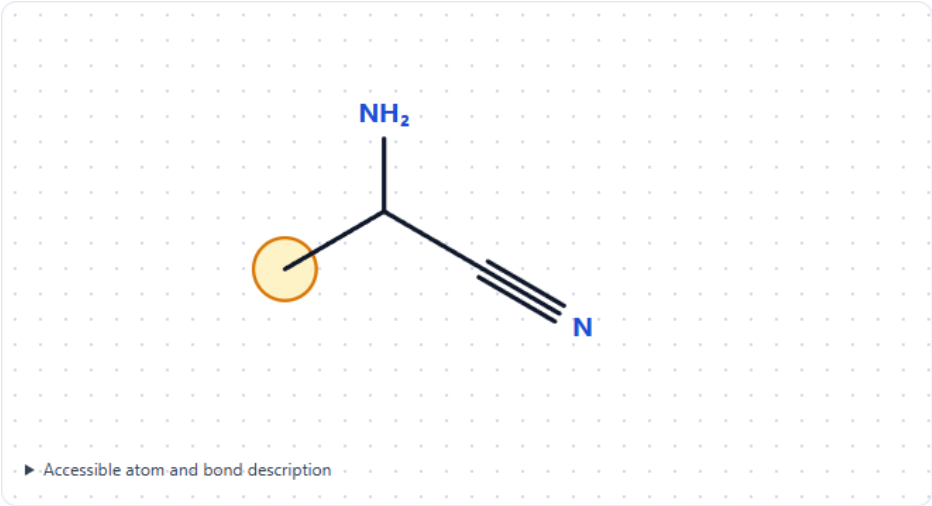
Answer type: choice

Option 0 (A): Racemic

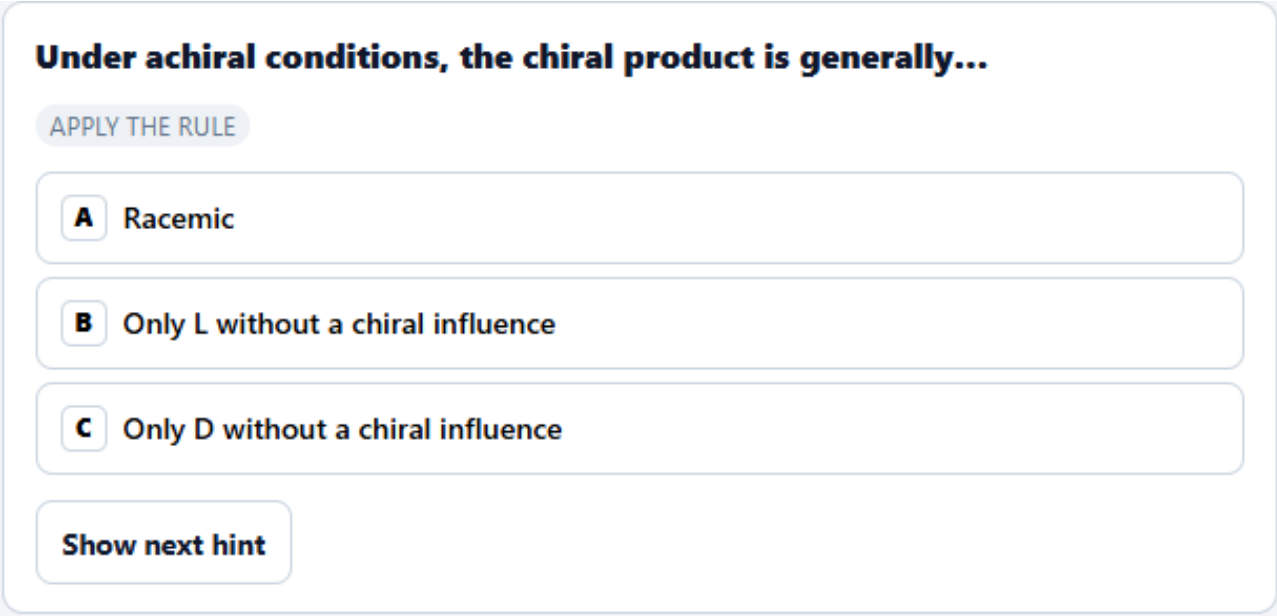
Option 1 (B): Only L without a chiral influence

Option 2 (C): Only D without a chiral influence

Screenshot: strecker-map-stage-3-molecule



Screenshot: strecker-map-stage-3-question



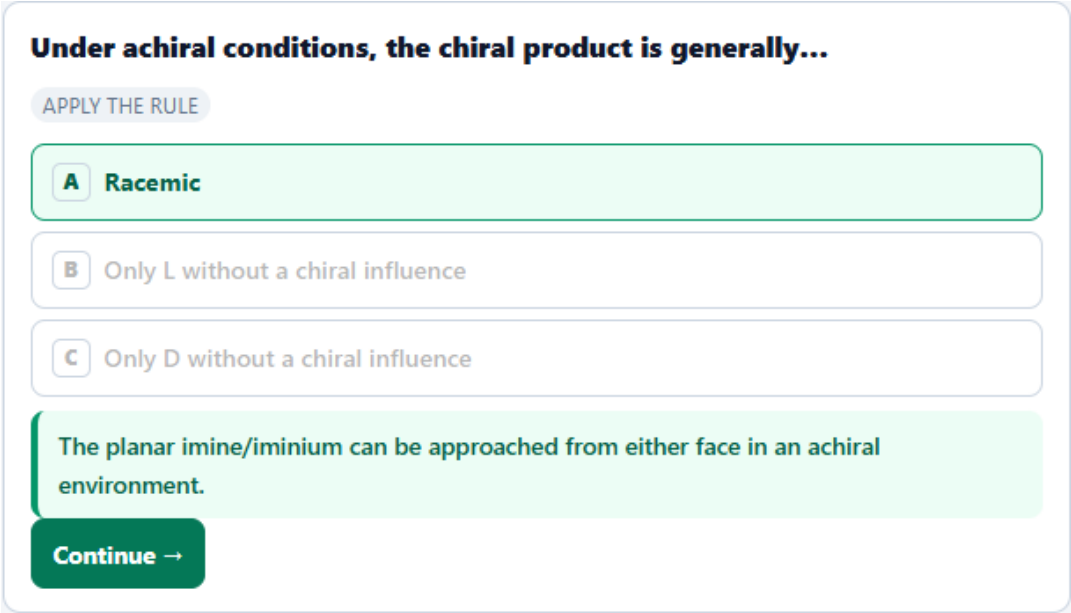
{"authorAnswer": 0, "tolerance": null, "highlightsAfterCorrectAnswer": {}}

Hint 1: Ask whether the reaction distinguishes enantiotopic faces.

Hint 2: No chiral catalyst or reagent is specified.

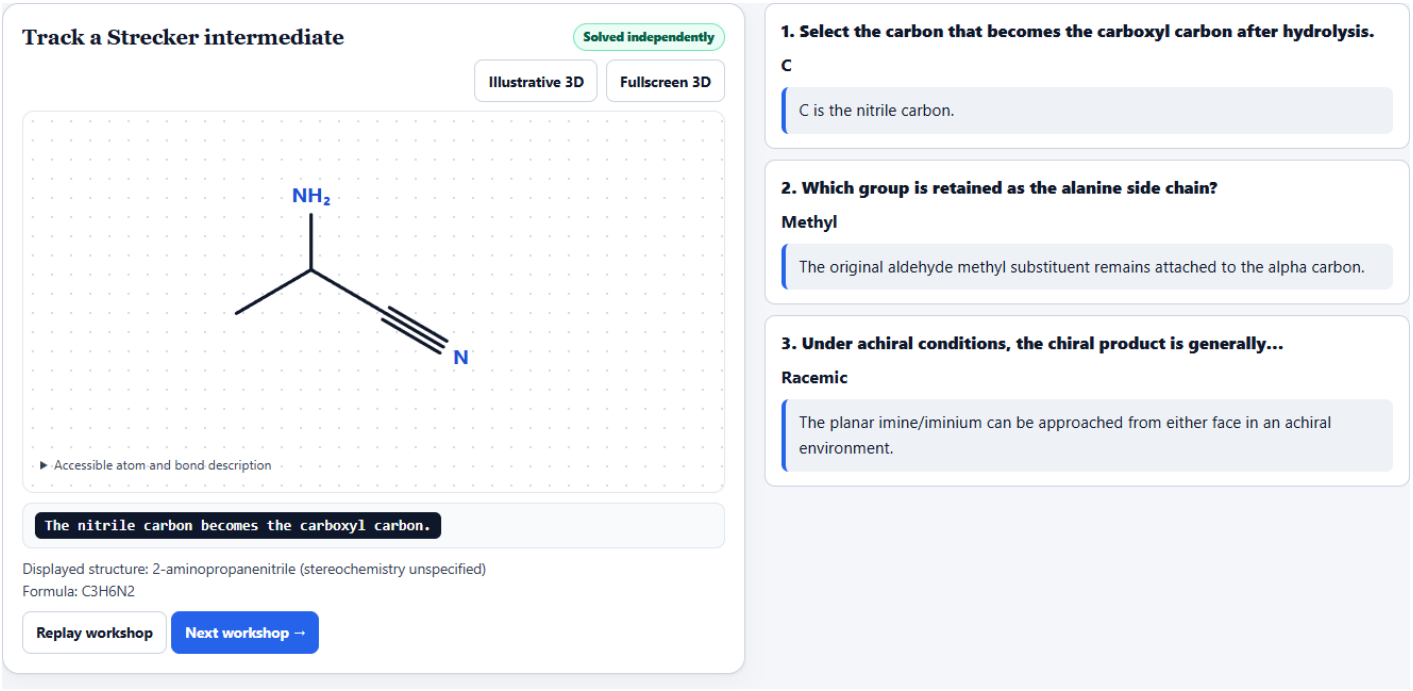
Explanation: The planar imine/iminium can be approached from either face in an achiral environment.

Screenshot: strecker-map-stage-3-feedback



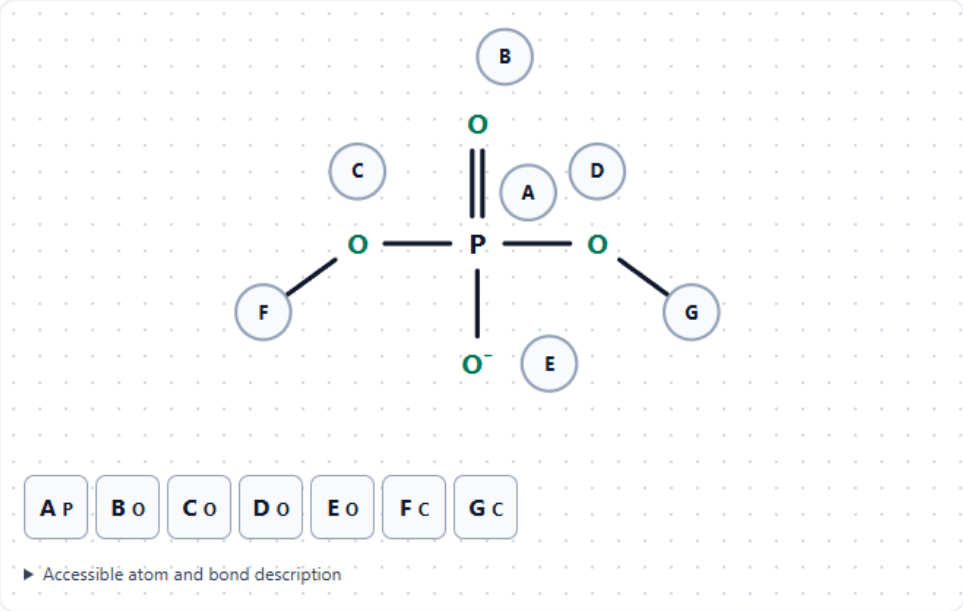
Completion summary: The nitrile carbon becomes the carboxyl carbon.

Screenshot: strecker-map-complete



phosphate-links | Identify phosphorus linkages

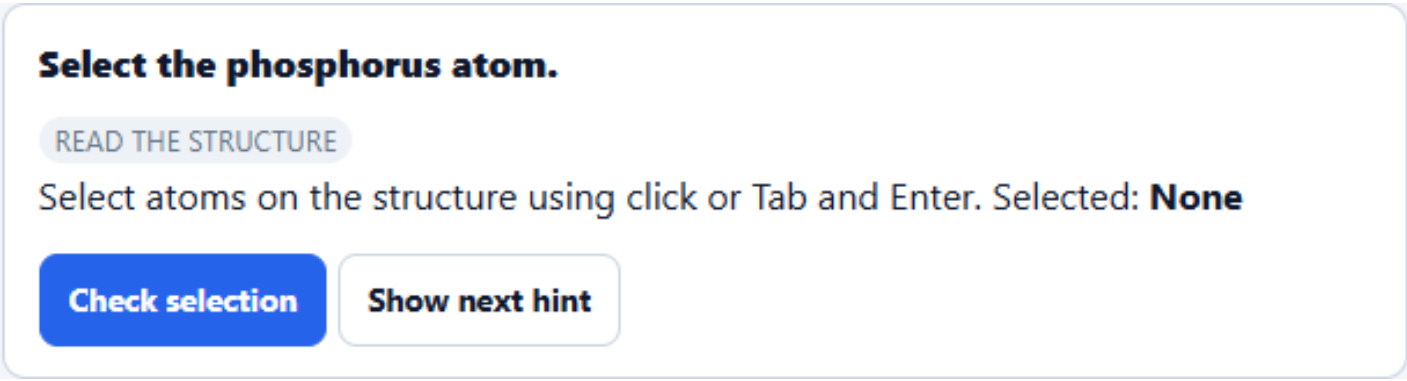
Screenshot: phosphate-links-stage-1-molecule



phosphate-links:stage-1 | Select the phosphorus atom.

Answer type: atoms

Screenshot: phosphate-links-stage-1-question



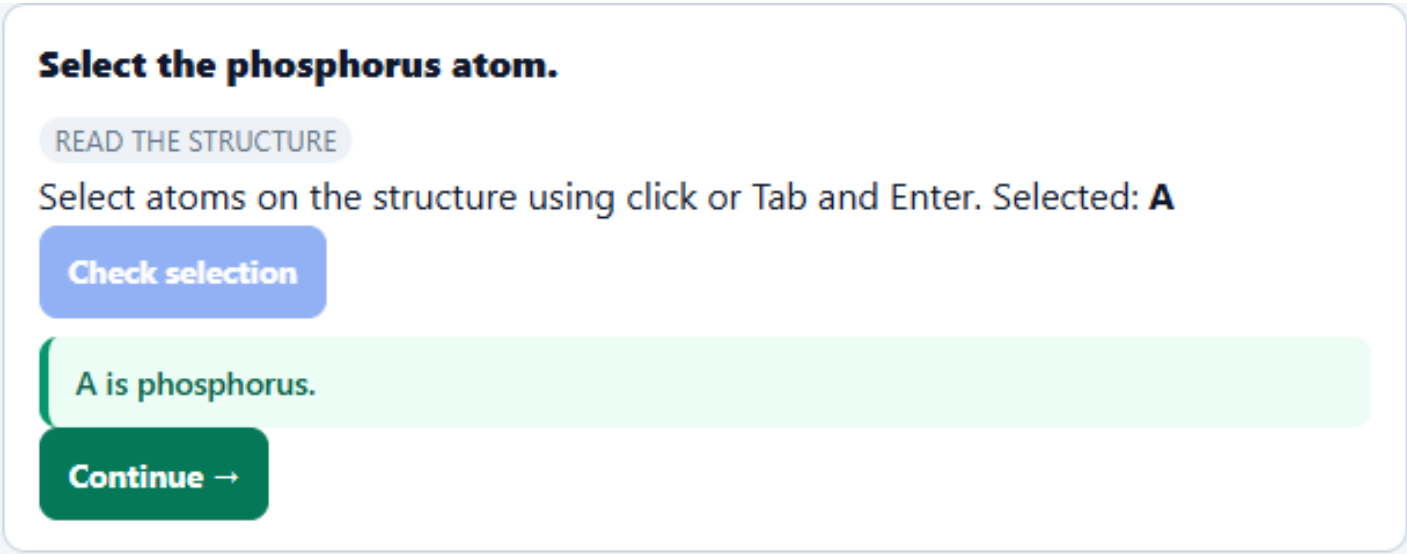
{"authorAnswer": [0], "tolerance": null, "highlightsAfterCorrectAnswer": {"prin": [0]}}

Hint 1: Read the P label.

Hint 2: It is bonded to four oxygens.

Explanation: A is phosphorus.

Screenshot: phosphate-links-stage-1-feedback



phosphate-links:stage-2 | This structure models a...

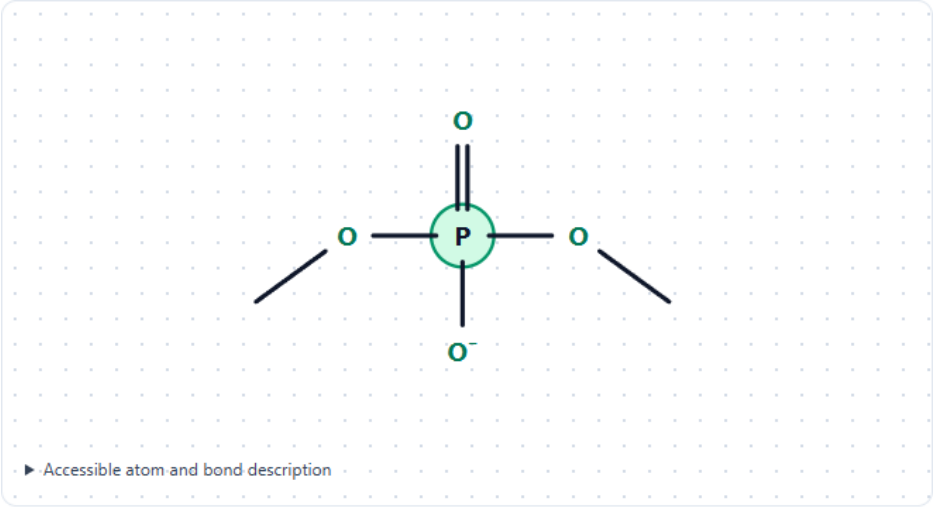
Answer type: choice

Option 0 (A): Phosphodiester

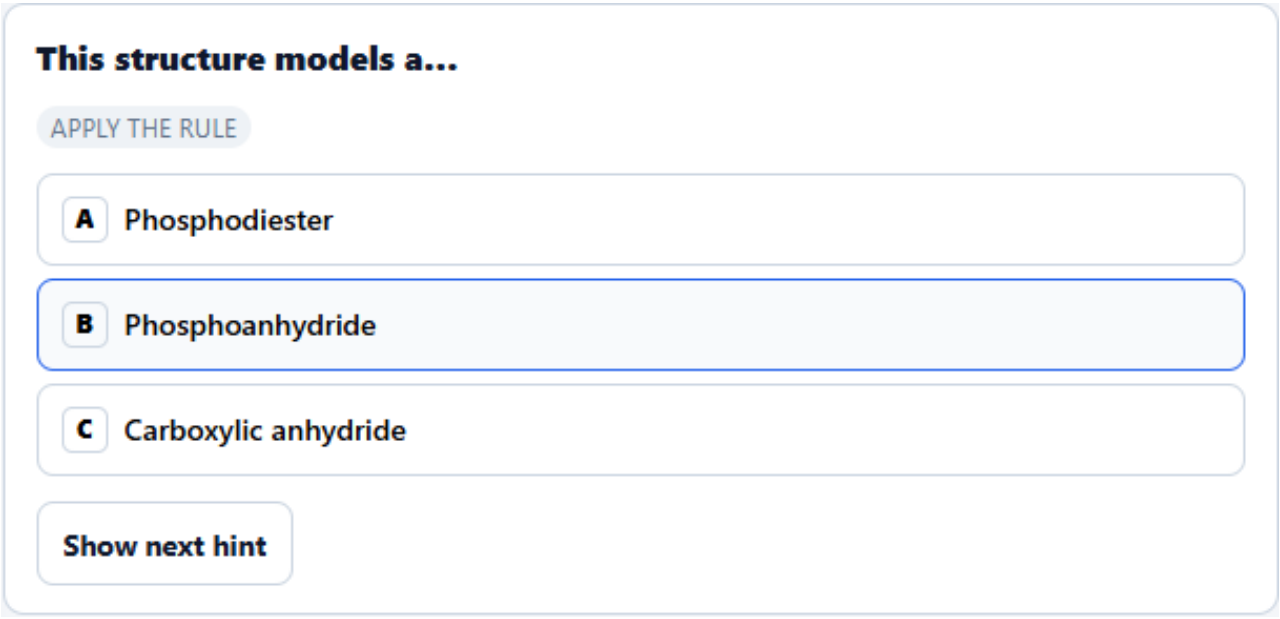
Option 1 (B): Phosphoanhydride

Option 2 (C): Carboxylic anhydride

Screenshot: phosphate-links-stage-2-molecule



Screenshot: phosphate-links-stage-2-question



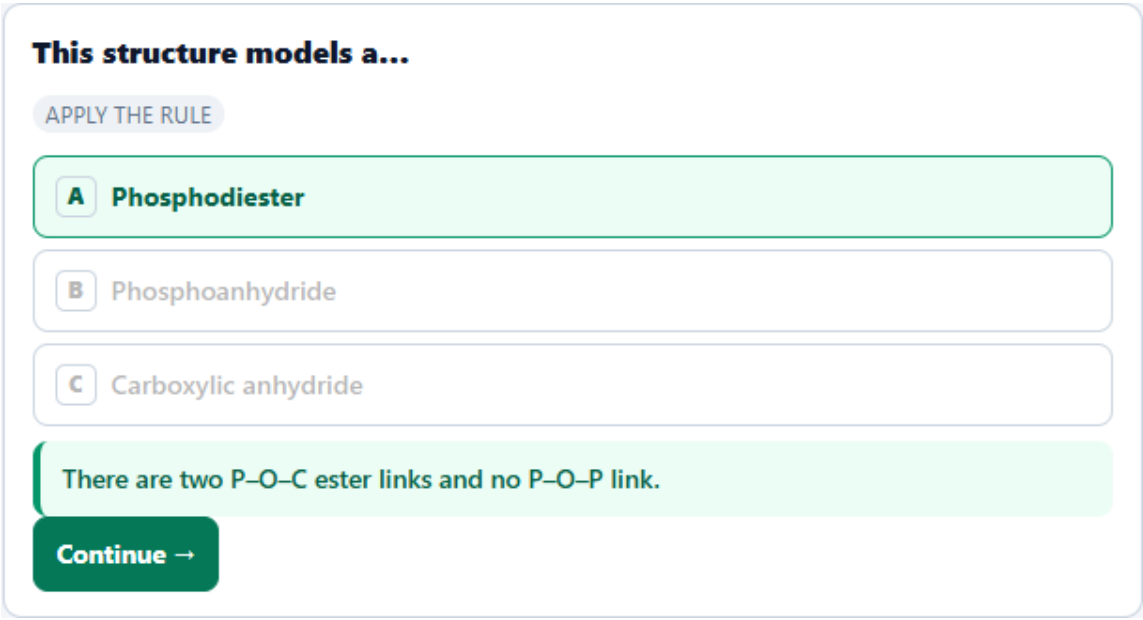
{"authorAnswer": 0, "tolerance": null, "highlightsAfterCorrectAnswer": {}}

Hint 1: Count oxygen links to carbon groups.

Hint 2: Only one phosphorus is present.

Explanation: There are two P–O–C ester links and no P–O–P link.

Screenshot: phosphate-links-stage-2-feedback



phosphate-links:stage-3 | Why can ATP hydrolysis be favorable overall?

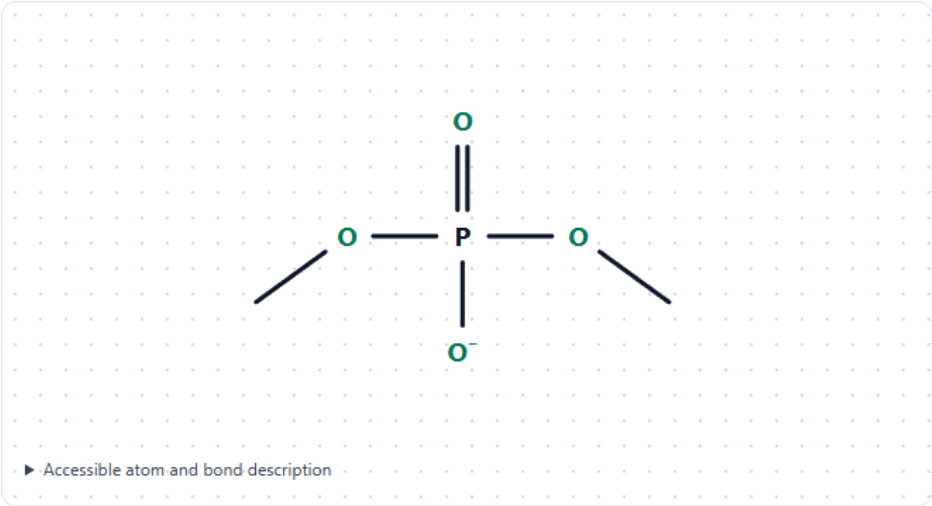
Answer type: choice

Option 0 (A): Products have lower free energy under the stated conditions

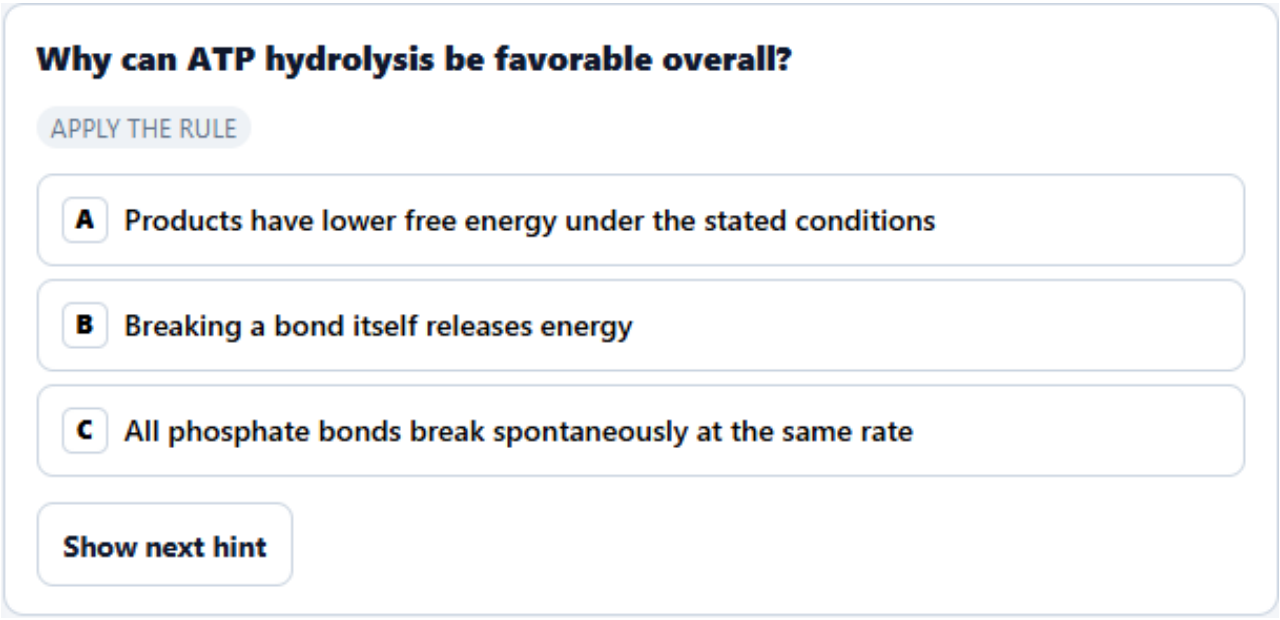
Option 1 (B): Breaking a bond itself releases energy

Option 2 (C): All phosphate bonds break spontaneously at the same rate

Screenshot: phosphate-links-stage-3-molecule



Screenshot: phosphate-links-stage-3-question



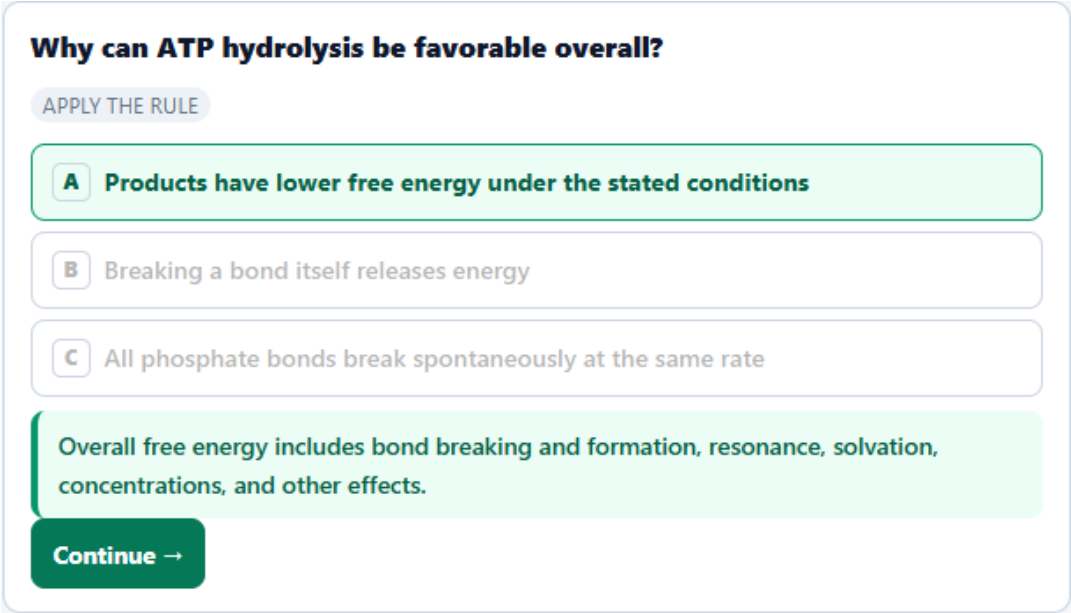
{"authorAnswer": 0, "tolerance": null, "highlightsAfterCorrectAnswer": {}}

Hint 1: Distinguish a single bond-breaking step from the full reaction.

Hint 2: Thermodynamics does not determine the rate by itself.

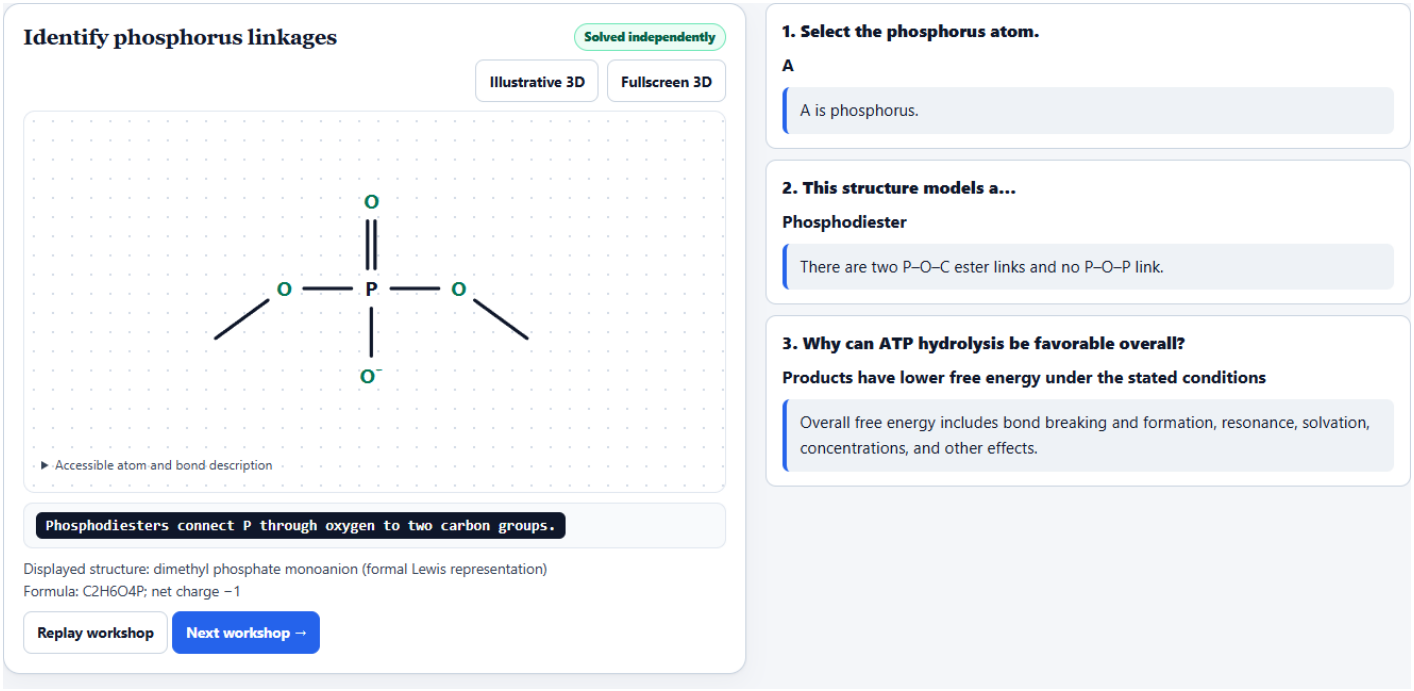
Explanation: Overall free energy includes bond breaking and formation, resonance, solvation, concentrations, and other effects.

Screenshot: phosphate-links-stage-3-feedback



Completion summary: Phosphodiesters connect P through oxygen to two carbon groups.

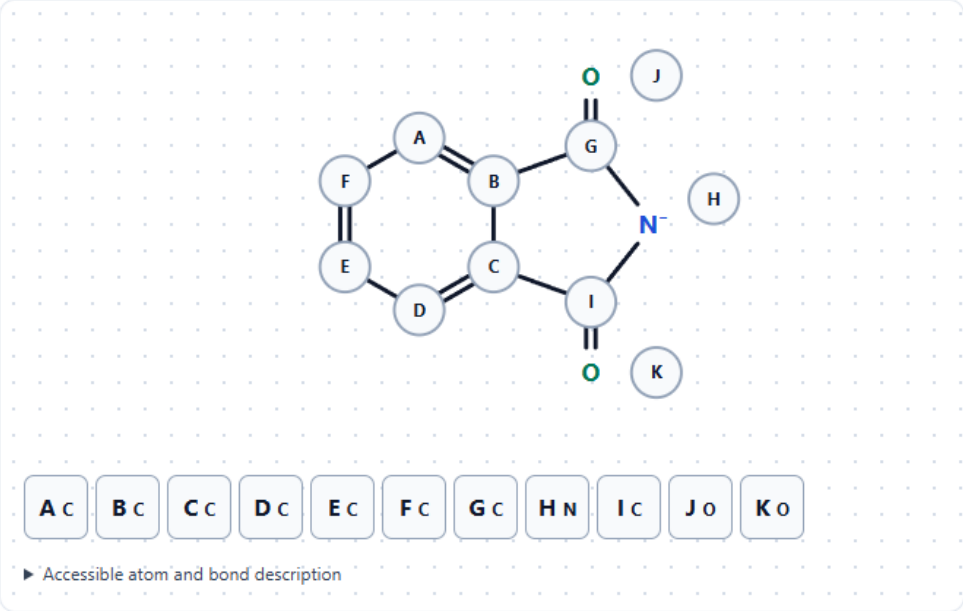
Screenshot: phosphate-links-complete



gabriel-route | Track protected nitrogen and side-chain installation

The canvas shows the protected nitrogen reagent, with its potassium counterion omitted; later questions refer to the route described in Learn & Explore.

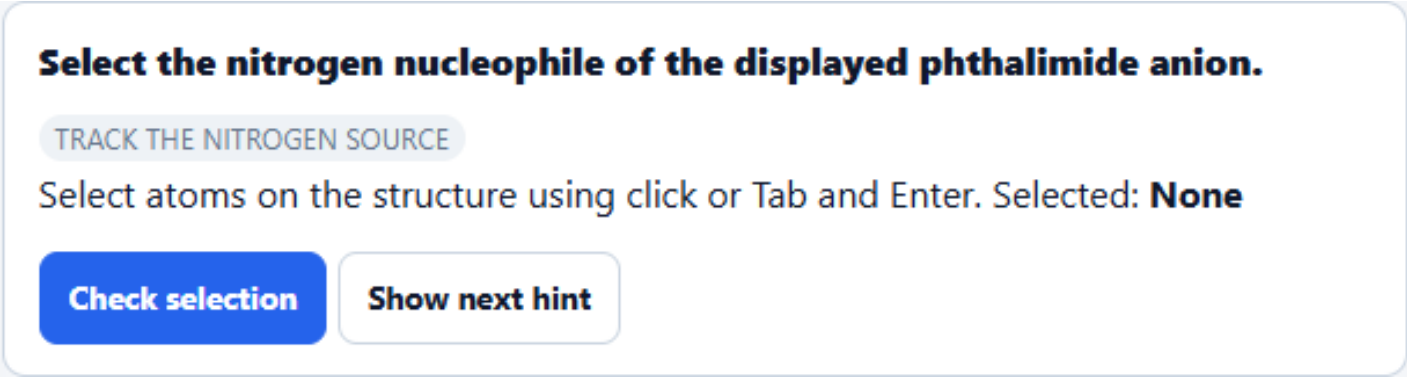
Screenshot: gabriel-route-stage-1-molecule



gabriel-route:stage-1 | Select the nitrogen nucleophile of the displayed phthalimide anion.

Answer type: atoms

Screenshot: gabriel-route-stage-1-question



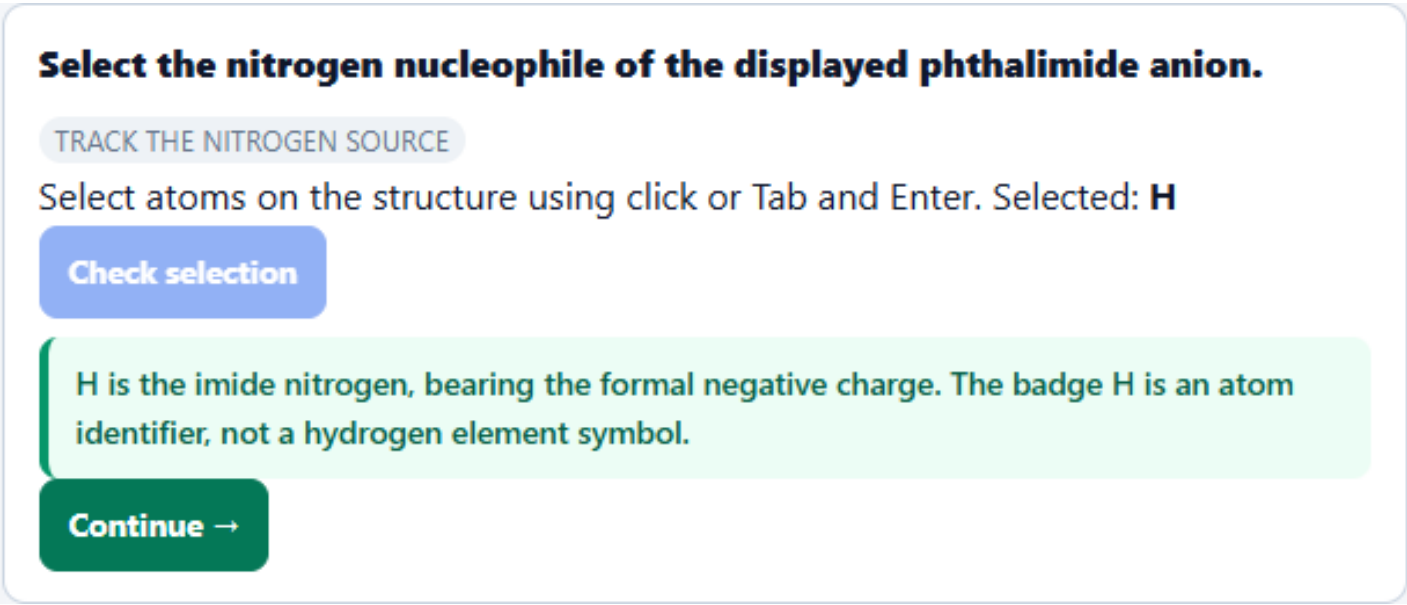
{"authorAnswer": [7], "tolerance": null, "highlightsAfterCorrectAnswer": {"prin": [7]}}

Hint 1: Locate N in the five-membered imide ring.

Hint 2: The nucleophilic atom bears the formal negative charge.

Explanation: H is the imide nitrogen, bearing the formal negative charge. The badge H is an atom identifier, not a hydrogen element symbol.

Screenshot: gabriel-route-stage-1-feedback



gabriel-route:stage-2 | After protected aminomalonate formation and deprotonation, a suitable primary alkyl halide installs the side chain mainly by...

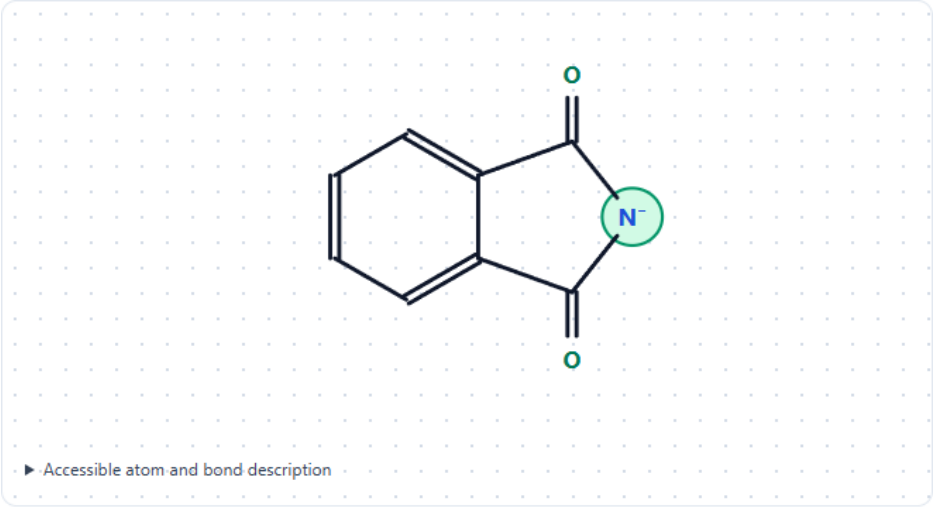
Answer type: choice

Option 0 (A): SN2 carbon alkylation

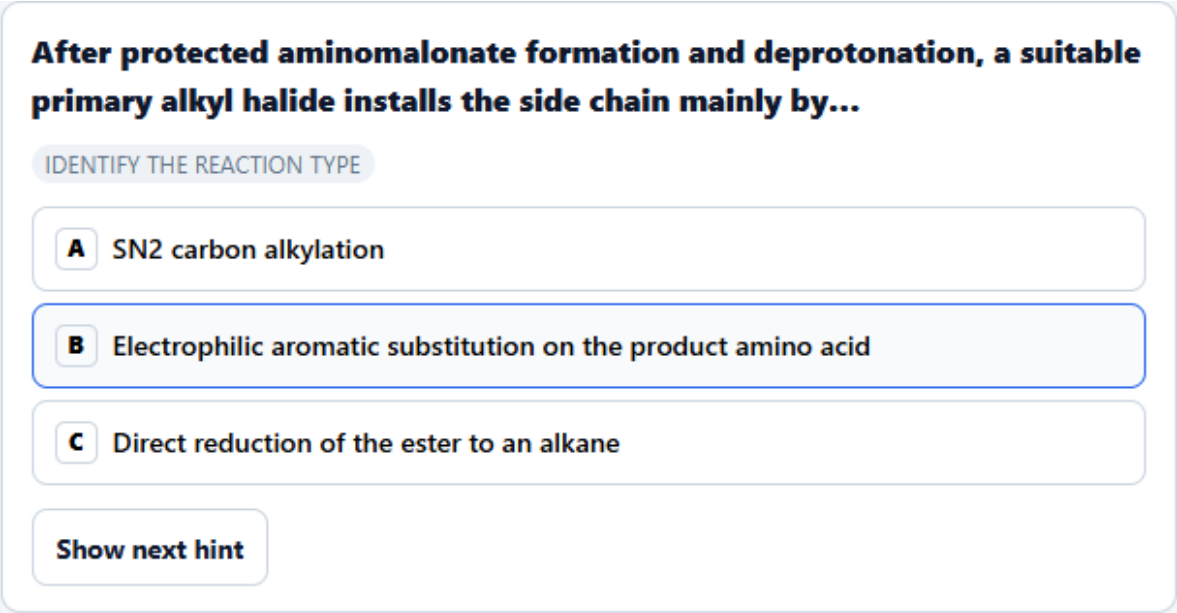
Option 1 (B): Electrophilic aromatic substitution on the product amino acid

Option 2 (C): Direct reduction of the ester to an alkane

Screenshot: gabriel-route-stage-2-molecule



Screenshot: gabriel-route-stage-2-question



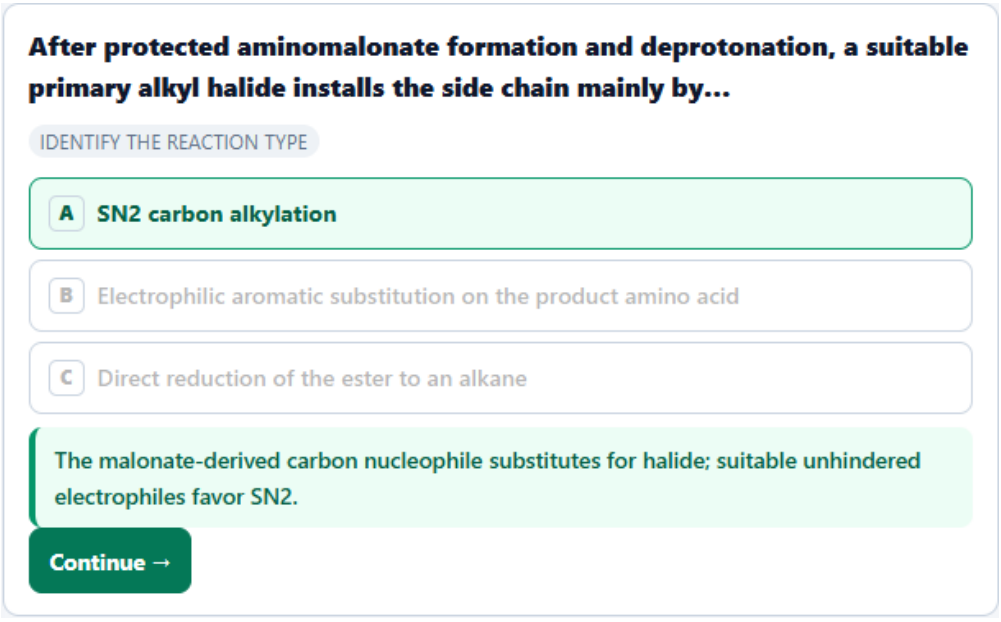
{"authorAnswer": 0, "tolerance": null, "highlightsAfterCorrectAnswer": {}}

Hint 1: Identify the nucleophile and the leaving group.

Hint 2: An alkyl halide is attacked at its saturated carbon.

Explanation: The malonate-derived carbon nucleophile substitutes for halide; suitable unhindered electrophiles favor SN2.

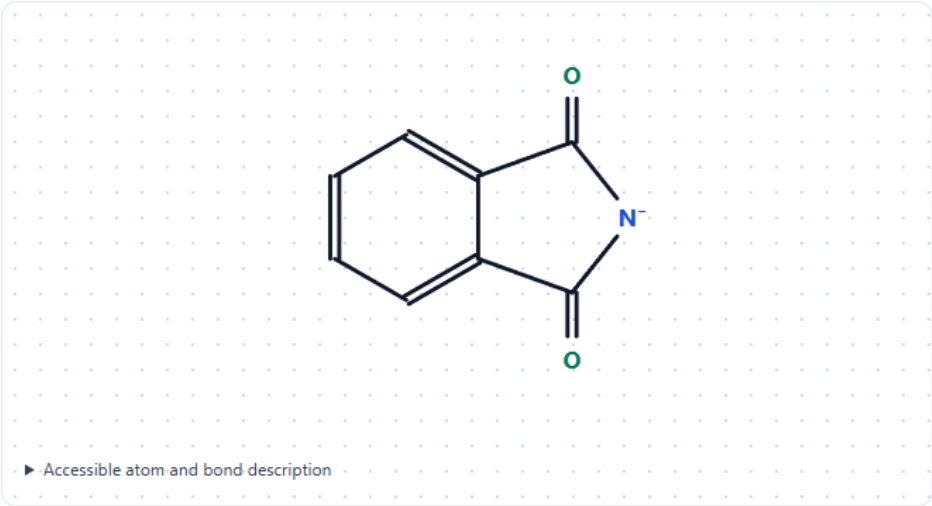
Screenshot: gabriel-route-stage-2-feedback



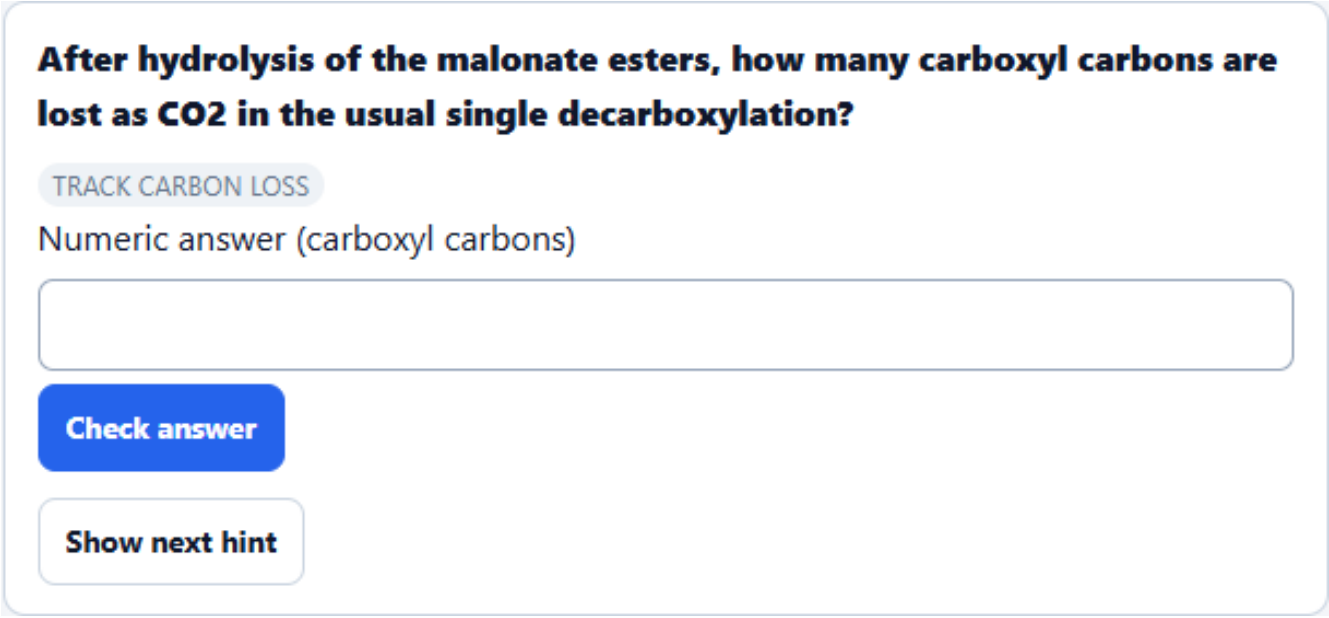
gabriel-route:stage-3 | After hydrolysis of the malonate esters, how many carboxyl carbons are lost as CO2 in the usual single decarboxylation?

Answer type: number | unit: carboxyl carbons

Screenshot: gabriel-route-stage-3-molecule



Screenshot: gabriel-route-stage-3-question



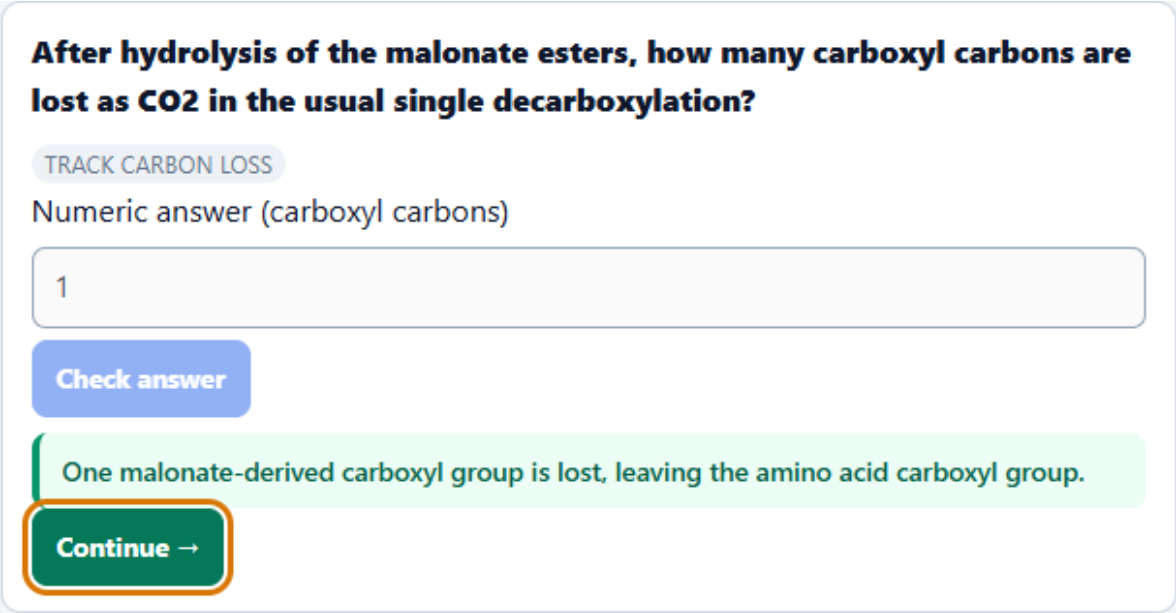
{"authorAnswer": 1, "tolerance": 0, "highlightsAfterCorrectAnswer": {}}

Hint 1: Distinguish deprotection from the decarboxylation step.

Hint 2: The aminomalonic acid framework starts with two carboxyl groups and retains one.

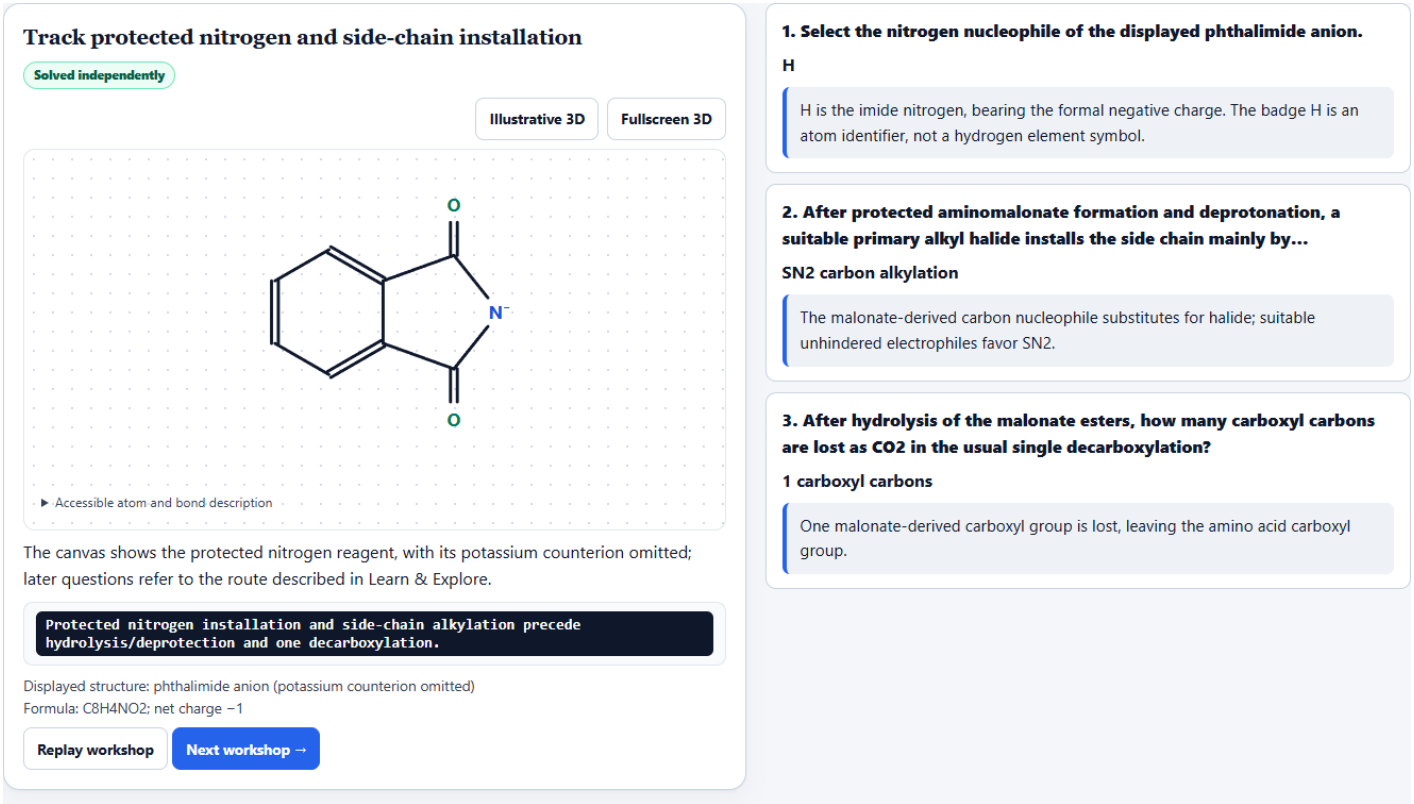
Explanation: One malonate-derived carboxyl group is lost, leaving the amino acid carboxyl group.

Screenshot: gabriel-route-stage-3-feedback



Completion summary: Protected nitrogen installation and side-chain alkylation precede hydrolysis/deprotection and one decarboxylation.

Screenshot: gabriel-route-complete



Molecular repository: author claims to verify

alanine

Author claims: alanine zwitterion (stereochemistry unspecified) | formula C3H7NO2 | canonical composition C3H7NO2 | net charge +0

Graph uses zero-based indices; A=0, B=1, etc. Coordinates are drawing positions, not 3D coordinates.

0 (A): N; x=45.00, y=115.00; charge=1; explicit H override=none

1 (B): C; x=105.00, y=85.00; charge=0; explicit H override=none

2 (C): C; x=165.00, y=115.00; charge=0; explicit H override=none

3 (D): O; x=165.00, y=175.00; charge=0; explicit H override=none

4 (E): O; x=225.00, y=85.00; charge=-1; explicit H override=none

5 (F): C; x=105.00, y=25.00; charge=0; explicit H override=none

Bonds [atom A, atom B, order]: [[0, 1, 1], [1, 2, 1], [2, 3, 2], [2, 4, 1], [1, 5, 1]]

glycine

Author claims: glycine zwitterion | formula C2H5NO2 | canonical composition C2H5NO2 | net charge +0

Graph uses zero-based indices; A=0, B=1, etc. Coordinates are drawing positions, not 3D coordinates.

0 (A): N; x=45.00, y=115.00; charge=1; explicit H override=none

1 (B): C; x=105.00, y=85.00; charge=0; explicit H override=none

2 (C): C; x=165.00, y=115.00; charge=0; explicit H override=none

3 (D): O; x=165.00, y=175.00; charge=0; explicit H override=none

4 (E): O; x=225.00, y=85.00; charge=-1; explicit H override=none

Bonds [atom A, atom B, order]: [[0, 1, 1], [1, 2, 1], [2, 3, 2], [2, 4, 1]]

glycylglycine

Author claims: glycylglycine zwitterion | formula C4H8N2O3 | canonical composition C4H8N2O3 | net charge +0

Graph uses zero-based indices; A=0, B=1, etc. Coordinates are drawing positions, not 3D coordinates.

0 (A): N; x=35.00, y=110.00; charge=1; explicit H override=none

1 (B): C; x=90.00, y=80.00; charge=0; explicit H override=none

2 (C): C; x=145.00, y=110.00; charge=0; explicit H override=none

3 (D): O; x=145.00, y=170.00; charge=0; explicit H override=none

4 (E): N; x=200.00, y=80.00; charge=0; explicit H override=none

5 (F): C; x=255.00, y=110.00; charge=0; explicit H override=none

6 (G): C; x=310.00, y=80.00; charge=0; explicit H override=none

7 (H): O; x=310.00, y=20.00; charge=0; explicit H override=none

8 (I): O; x=365.00, y=110.00; charge=-1; explicit H override=none

Bonds [atom A, atom B, order]: [[0, 1, 1], [1, 2, 1], [2, 3, 2], [2, 4, 1], [4, 5, 1], [5, 6, 1], [6, 7, 2], [6, 8, 1]]

aminonitrile

Author claims: 2-aminopropanenitrile (stereochemistry unspecified) | formula C3H6N2 | canonical composition C3H6N2 | net charge +0

Graph uses zero-based indices; A=0, B=1, etc. Coordinates are drawing positions, not 3D coordinates.

0 (A): C; x=50.00, y=120.00; charge=0; explicit H override=none

1 (B): C; x=110.00, y=85.00; charge=0; explicit H override=none

2 (C): C; x=170.00, y=120.00; charge=0; explicit H override=none

3 (D): N; x=230.00, y=155.00; charge=0; explicit H override=none

4 (E): N; x=110.00, y=25.00; charge=0; explicit H override=none

Bonds [atom A, atom B, order]: [[0, 1, 1], [1, 2, 1], [2, 3, 3], [1, 4, 1]]

phthalimide-anion

Author claims: phthalimide anion (potassium counterion omitted) | formula C8H4NO2 | canonical composition C8H4NO2 | net charge -1

Graph uses zero-based indices; A=0, B=1, etc. Coordinates are drawing positions, not 3D coordinates.

0 (A): C; x=130.00, y=45.00; charge=0; explicit H override=none

1 (B): C; x=177.63, y=72.50; charge=0; explicit H override=none

2 (C): C; x=177.63, y=127.50; charge=0; explicit H override=none

3 (D): C; x=130.00, y=155.00; charge=0; explicit H override=none

4 (E): C; x=82.37, y=127.50; charge=0; explicit H override=none

5 (F): C; x=82.37, y=72.50; charge=0; explicit H override=none

6 (G): C; x=240.00, y=50.00; charge=0; explicit H override=none

7 (H): N; x=280.00, y=100.00; charge=-1; explicit H override=0

8 (I): C; x=240.00, y=150.00; charge=0; explicit H override=none

9 (J): O; x=240.00, y=5.00; charge=0; explicit H override=none

10 (K): O; x=240.00, y=195.00; charge=0; explicit H override=none

Bonds [atom A, atom B, order]: [[0, 1, 2], [1, 2, 1], [2, 3, 2], [3, 4, 1], [4, 5, 2], [5, 0, 1], [1, 6, 1], [6, 7, 1], [7, 8, 1], [8, 2, 1], [6, 9, 2], [8, 10, 2]]

dimethylphosphate

Author claims: dimethyl phosphate monoanion (formal Lewis representation) | formula C2H6O4P | canonical composition C2H6O4P | net charge -1

Graph uses zero-based indices; A=0, B=1, etc. Coordinates are drawing positions, not 3D coordinates.

0 (A): P; x=160.00, y=100.00; charge=0; explicit H override=none

1 (B): O; x=160.00, y=30.00; charge=0; explicit H override=none

2 (C): O; x=90.00, y=100.00; charge=0; explicit H override=none

3 (D): O; x=230.00, y=100.00; charge=0; explicit H override=none

4 (E): O; x=160.00, y=170.00; charge=-1; explicit H override=none

5 (F): C; x=35.00, y=140.00; charge=0; explicit H override=none

6 (G): C; x=285.00, y=140.00; charge=0; explicit H override=none

Bonds [atom A, atom B, order]: [[0, 1, 2], [0, 2, 1], [0, 3, 1], [0, 4, 1], [2, 5, 1], [3, 6, 1]]

phosphoricacid

Author claims: phosphoric acid (formal Lewis representation) | formula H3PO4 | canonical composition H3O4P | net charge +0

Graph uses zero-based indices; A=0, B=1, etc. Coordinates are drawing positions, not 3D coordinates.

0 (A): P; x=150.00, y=100.00; charge=0; explicit H override=none

1 (B): O; x=150.00, y=30.00; charge=0; explicit H override=none

2 (C): O; x=80.00, y=100.00; charge=0; explicit H override=none

3 (D): O; x=220.00, y=100.00; charge=0; explicit H override=none

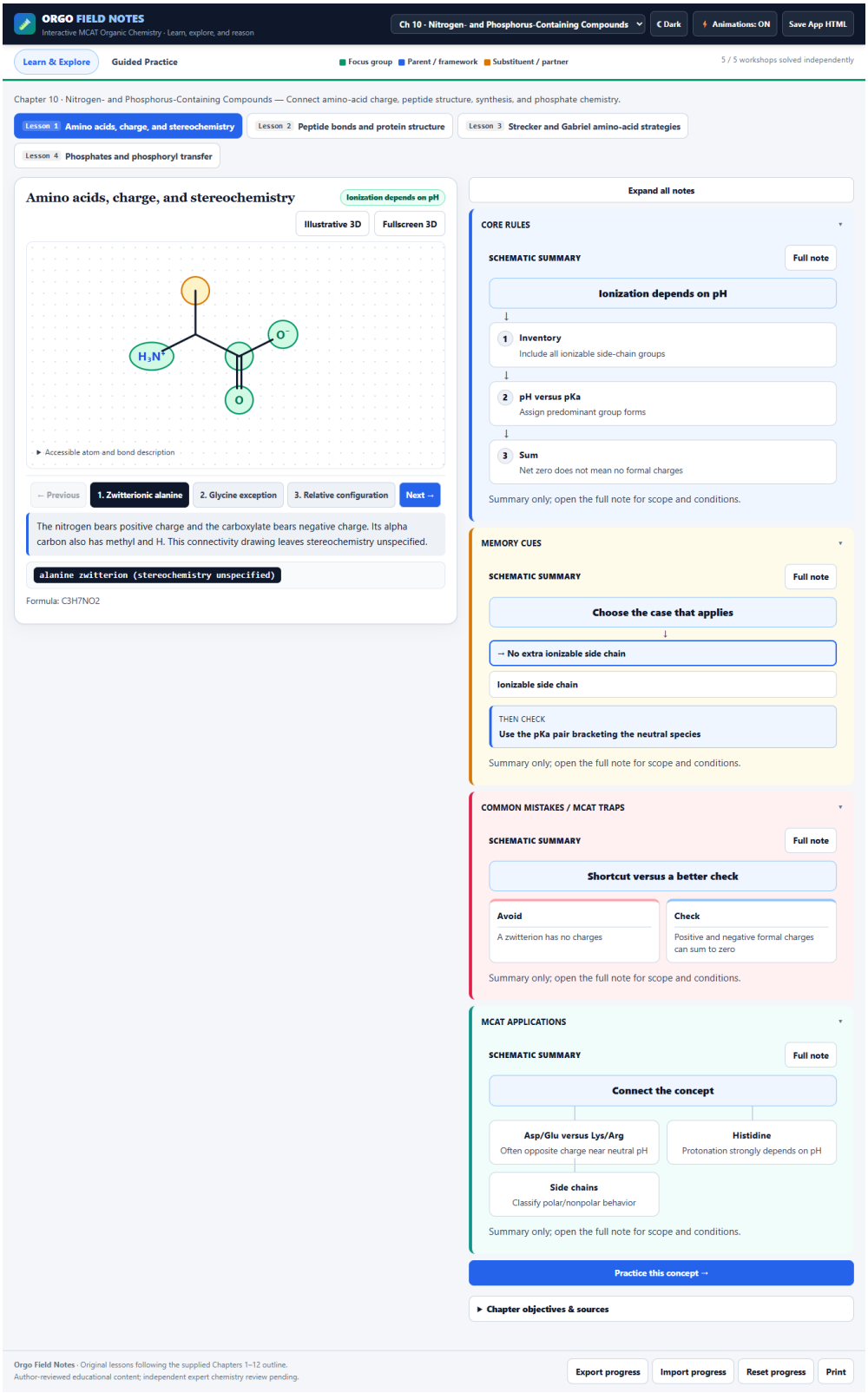
4 (E): O; x=150.00, y=170.00; charge=0; explicit H override=none

Bonds [atom A, atom B, order]: [[0, 1, 2], [0, 2, 1], [0, 3, 1], [0, 4, 1]]

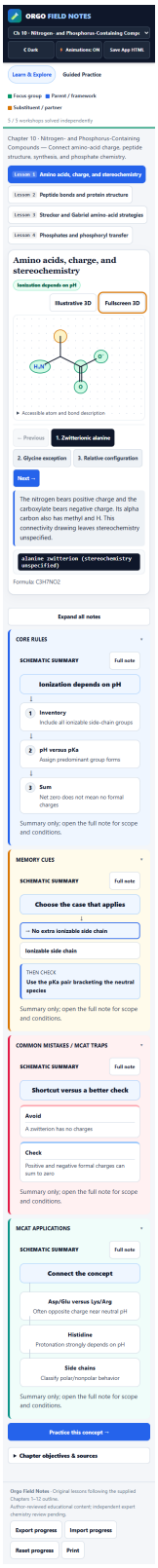
Layout samples

These overview images are scaled for page fit. Detailed structure and stage images appear earlier. Samples cover the first concept on desktop and mobile in both themes, not every responsive state.

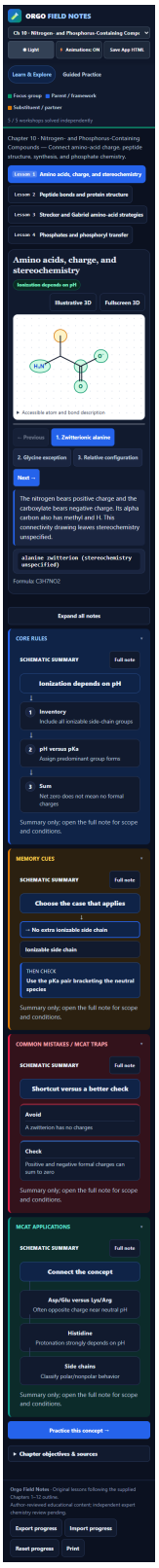
Screenshot: layout-desktop-light



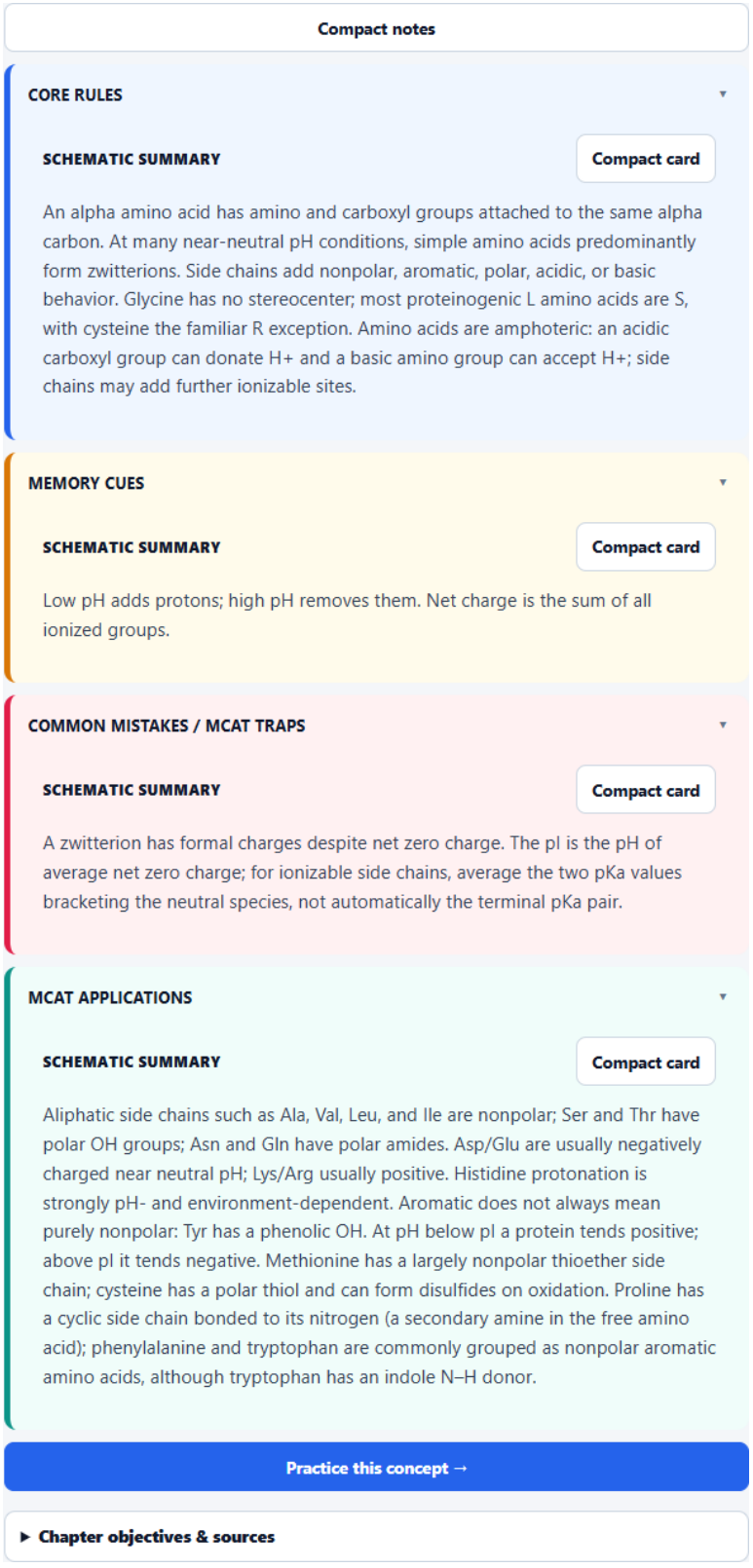
Screenshot: layout-mobile-light



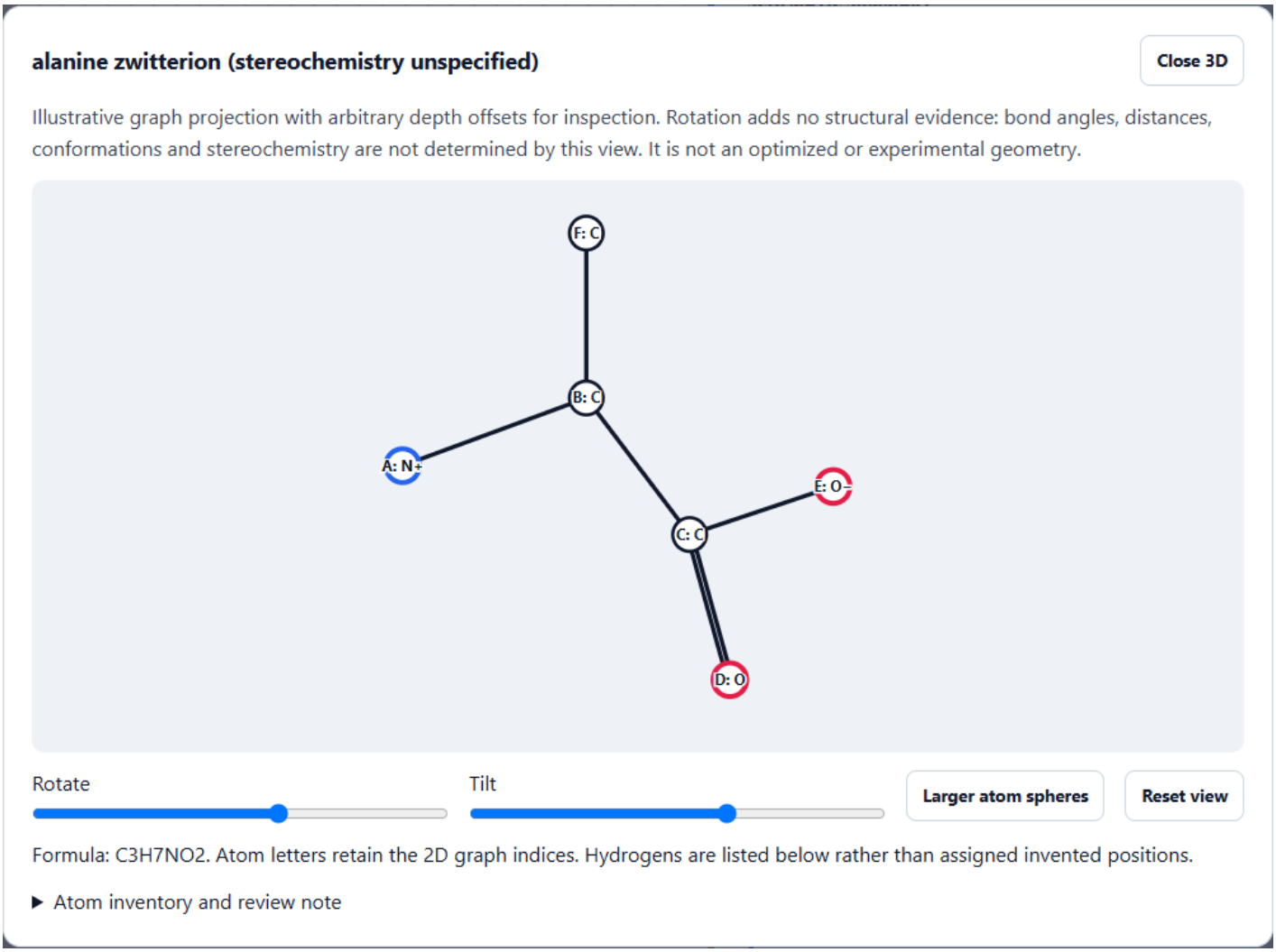
Screenshot: layout-mobile-dark



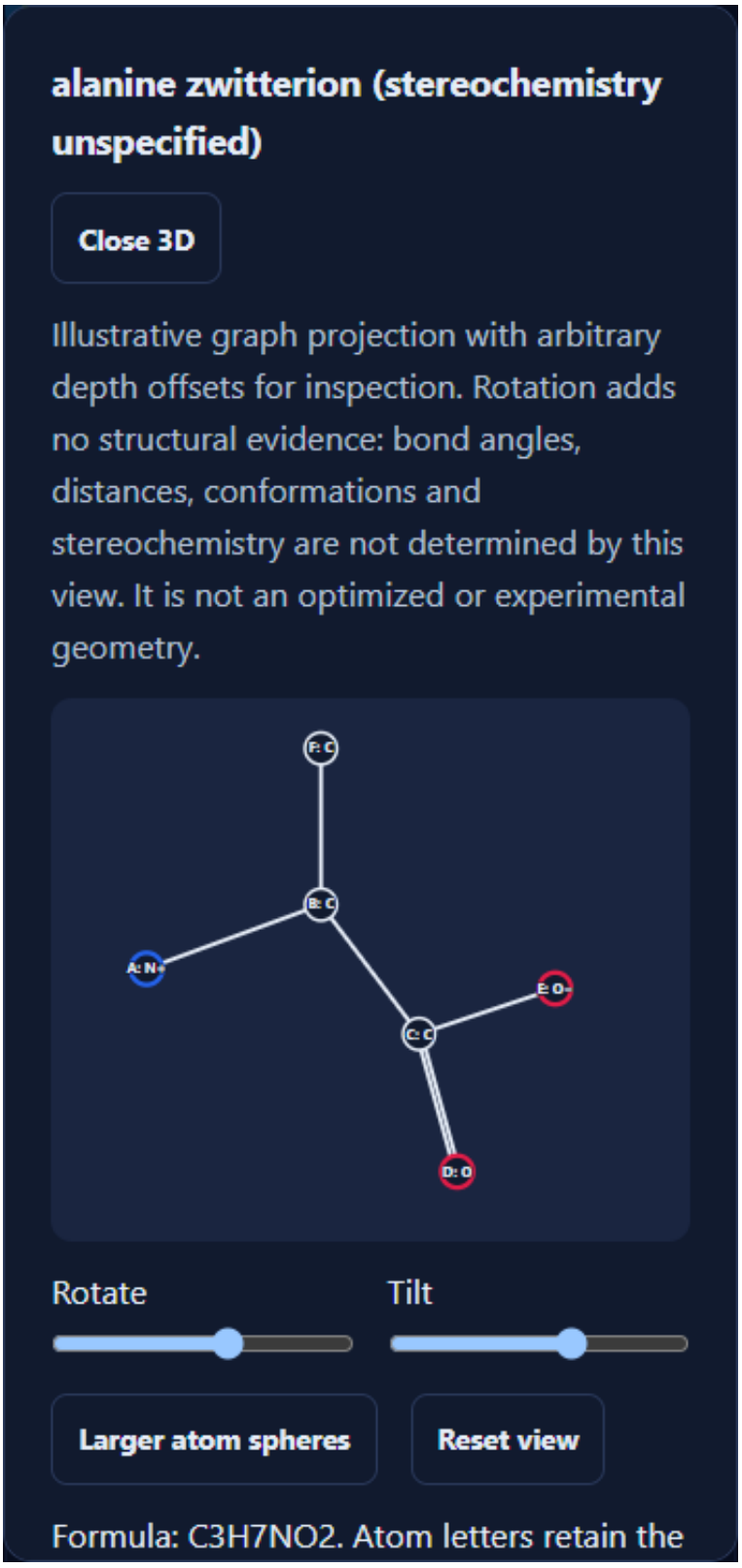
Screenshot: visual-notes-expanded



Screenshot: projection-desktop-light



Screenshot: projection-mobile-dark



Execution evidence

72 screenshots and 16 recorded interactions. Incorrect answers rejected and encoded keys accepted for captured guided stages. This checks implementation against its data; it does not validate chemistry independently. Static screenshots do not verify animation timing, arbitrary clicks, or screen-reader experience.